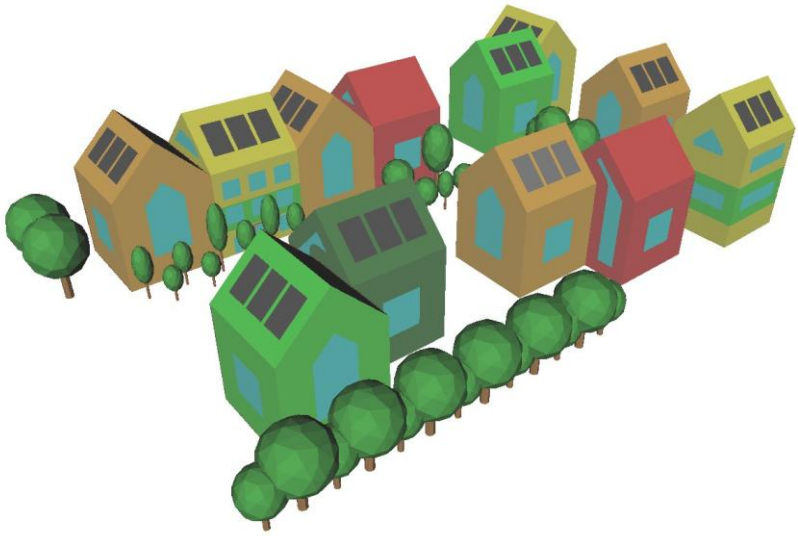
CityGML

energyADE 3.0

Beta 8

UML diagrams



Last updated on 16 October 2026

DigiTwins4PEDs

FlexPED

RENODAT

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The latest version of this document can be downloaded from:
https://github.com/tudelft3d/Energy_ADE/tree/3.0beta8/documentation

Author
Giorgio Agugiaro
3D Geoinformation Group
Department of Urbanism
Faculty of Architecture and Built Environment
Delft University of Technology
The Netherlands
g.agugiaro@tudelft.nl
https://3d.bk.tudelft.nl/gagugiaro/
https://3d.bk.tudelft.nl/urbanrg/

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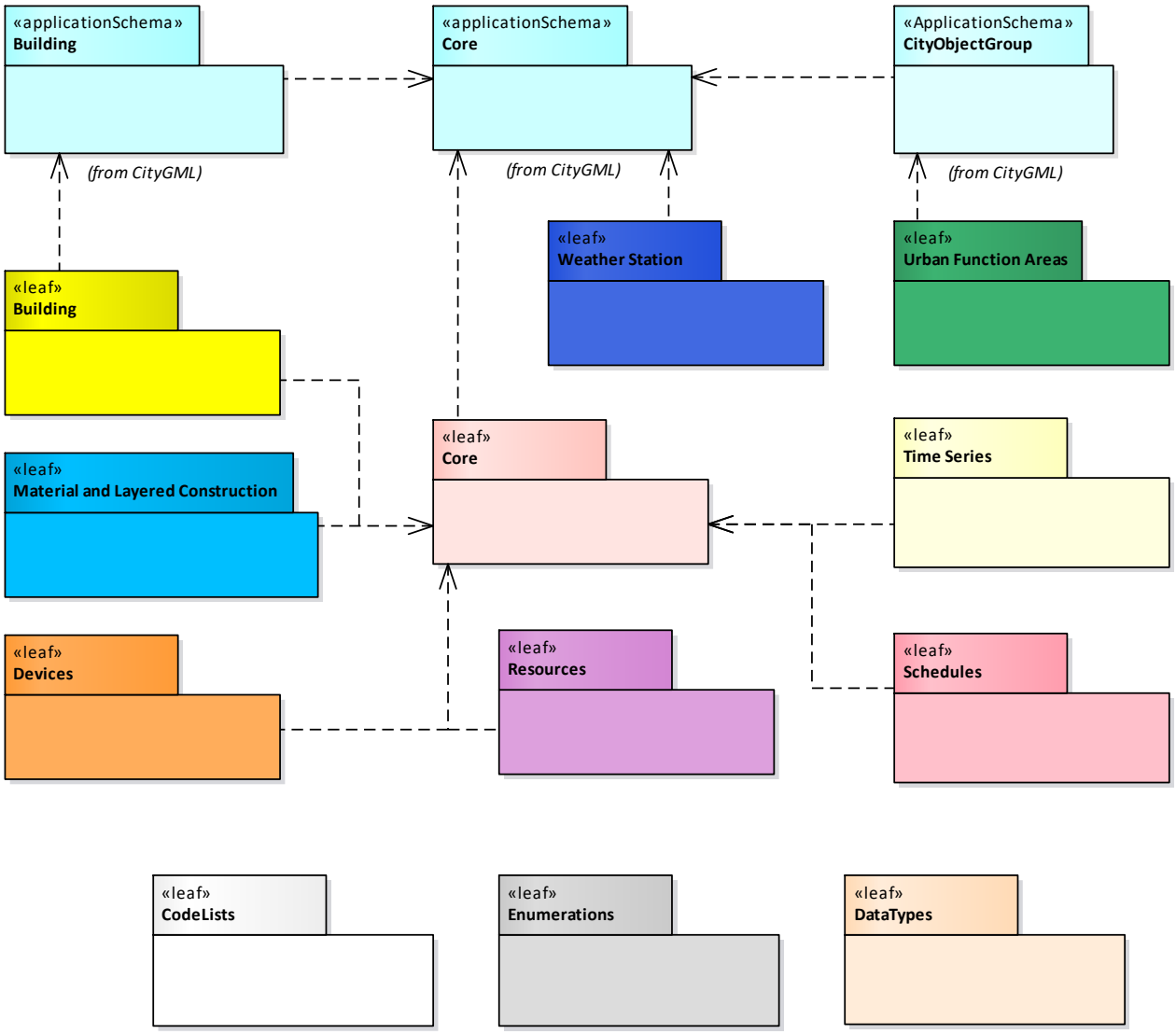
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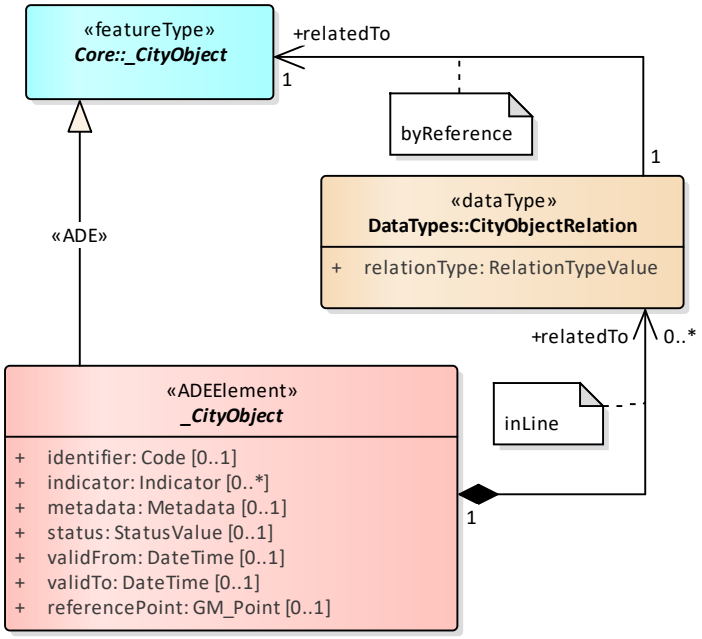
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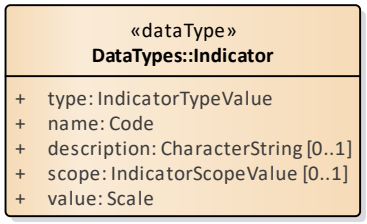
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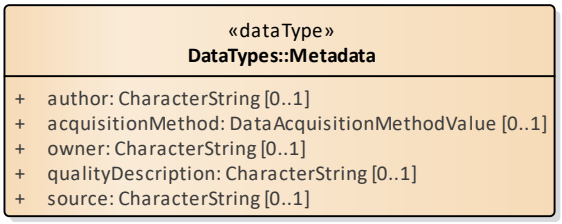
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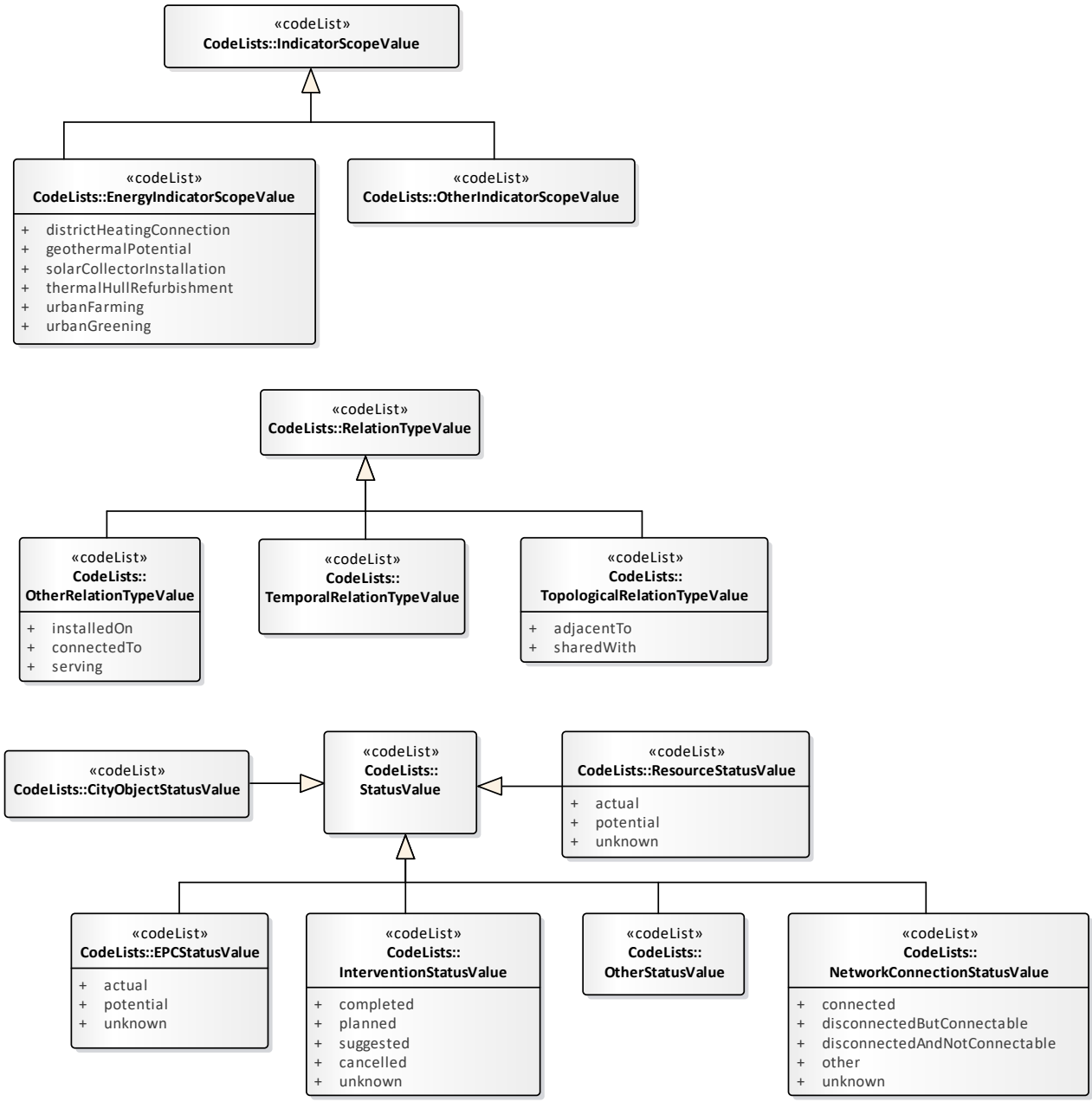
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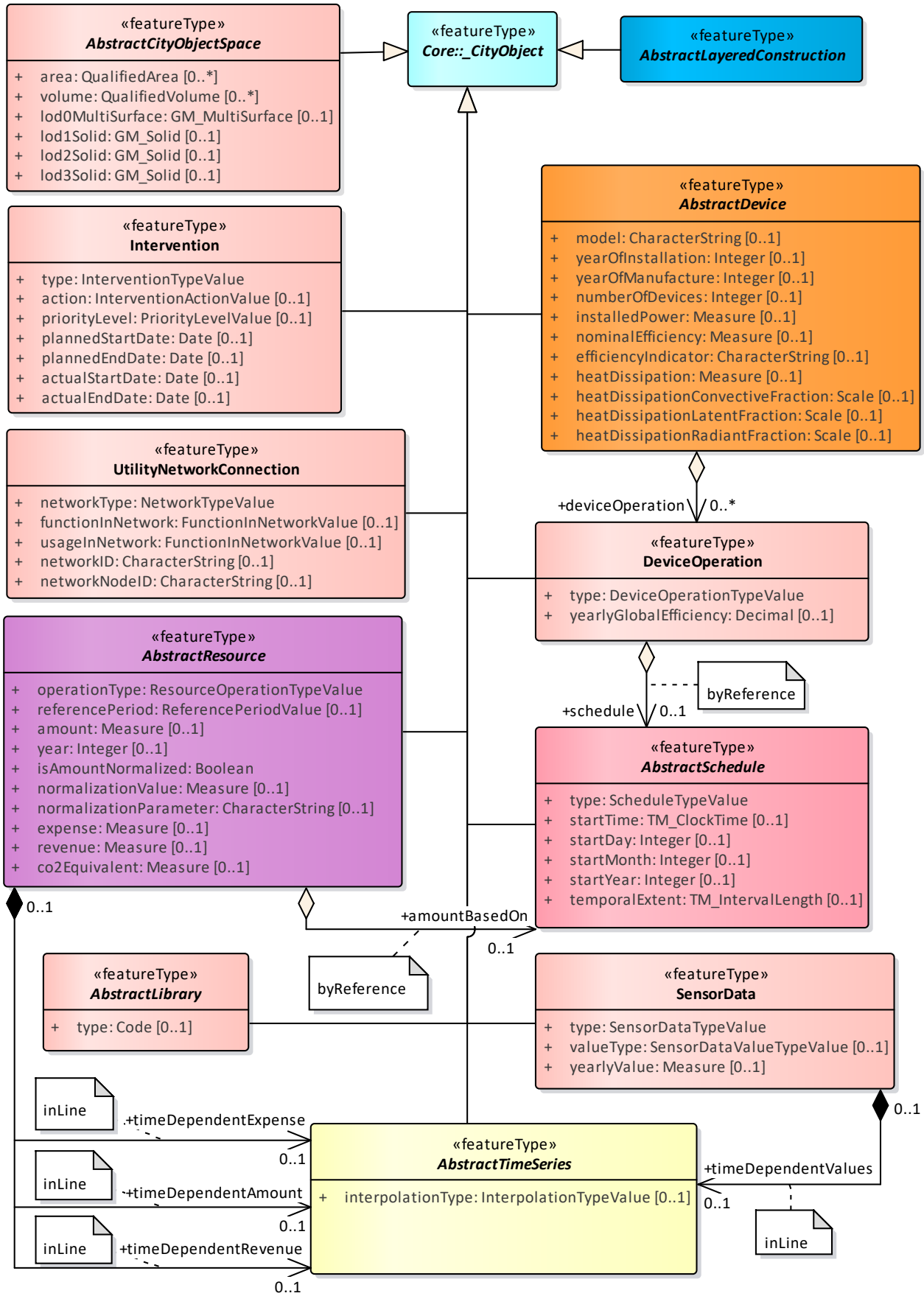




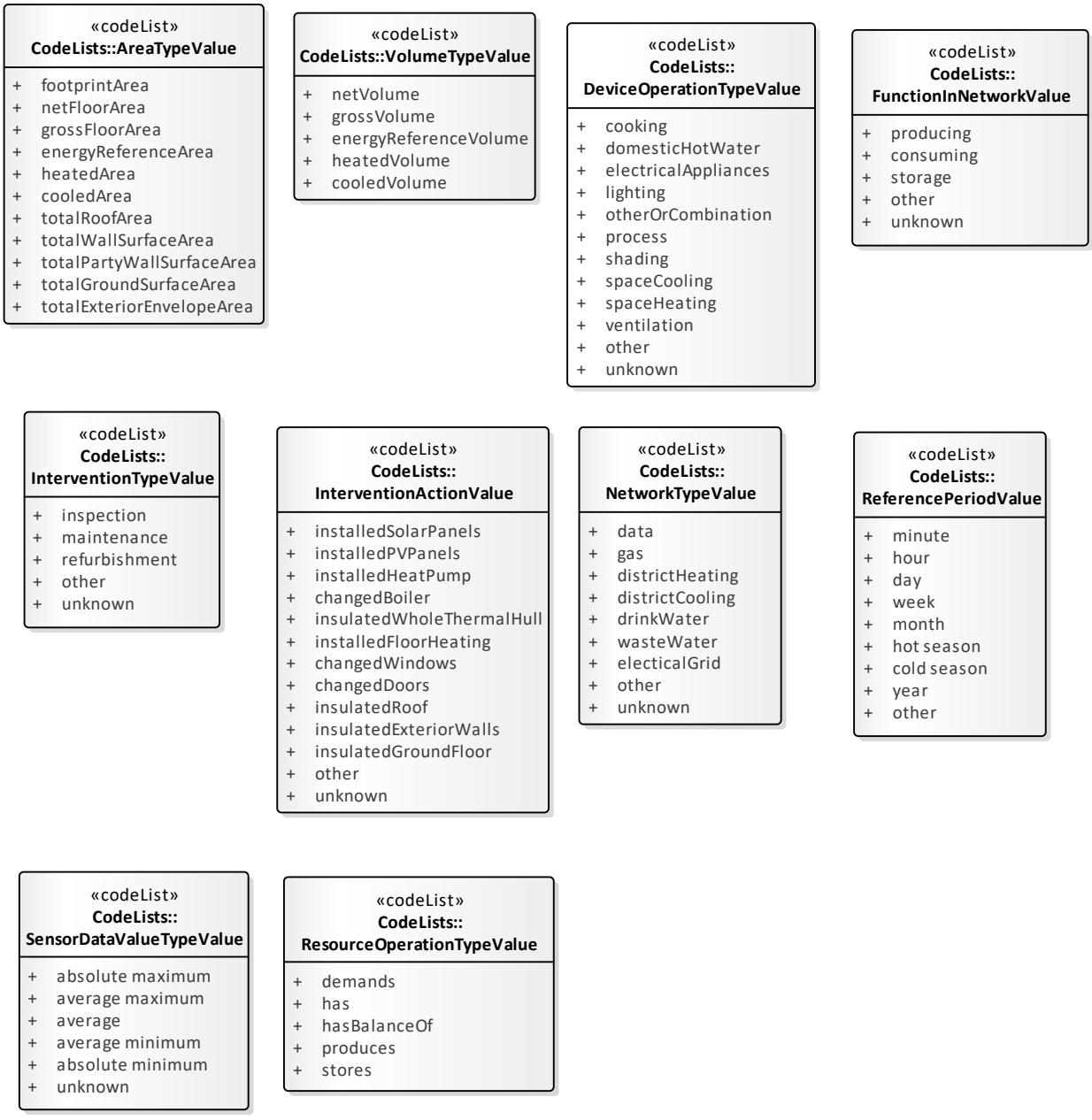
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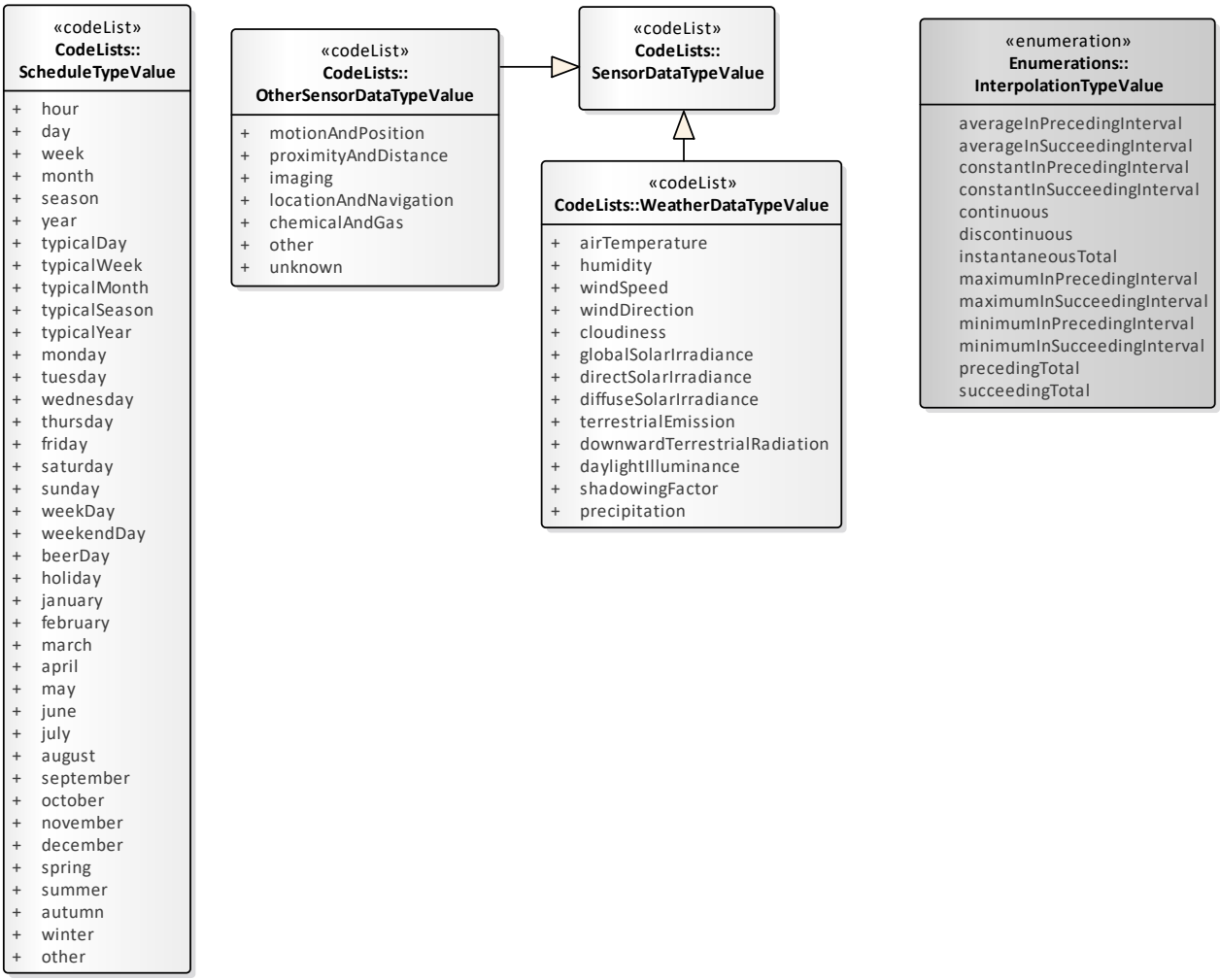
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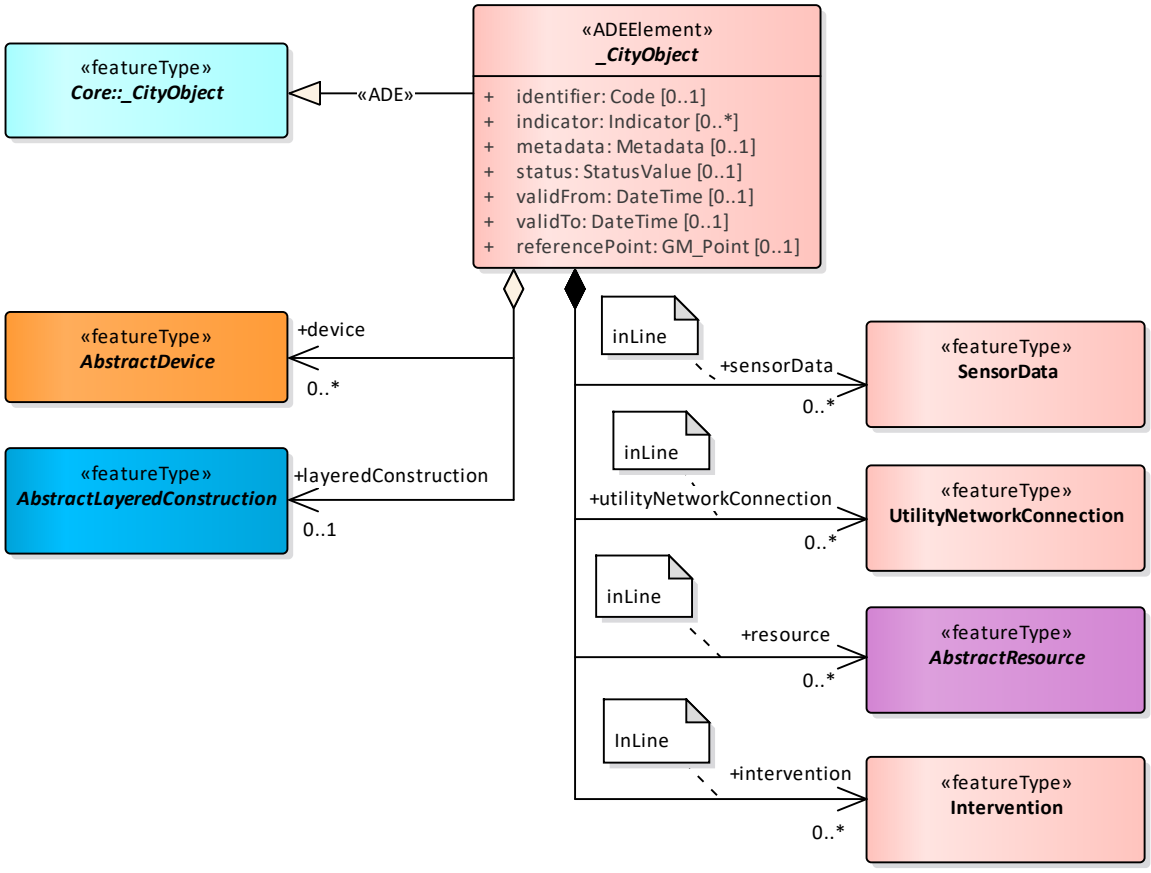
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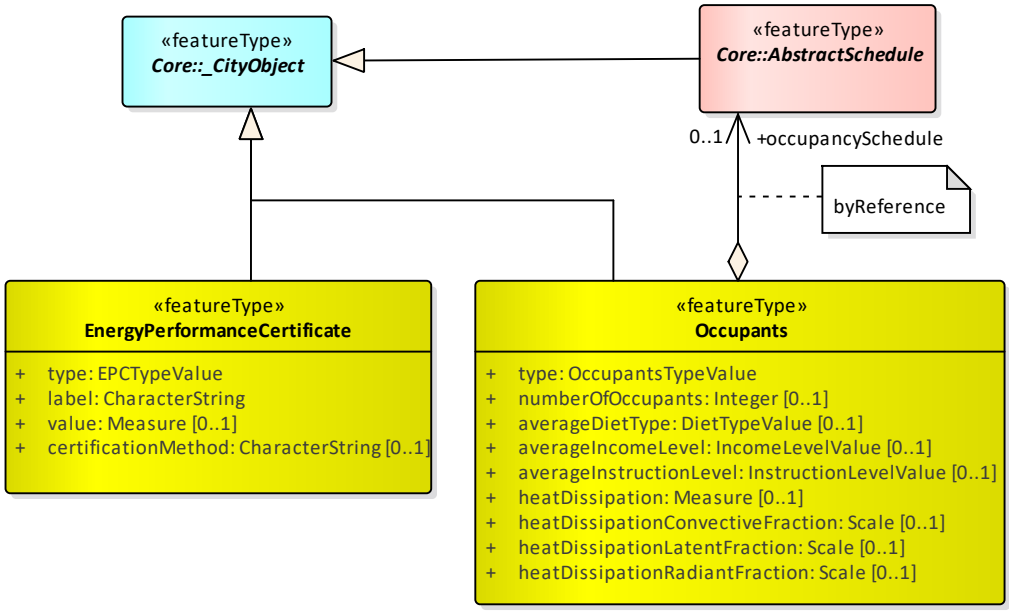
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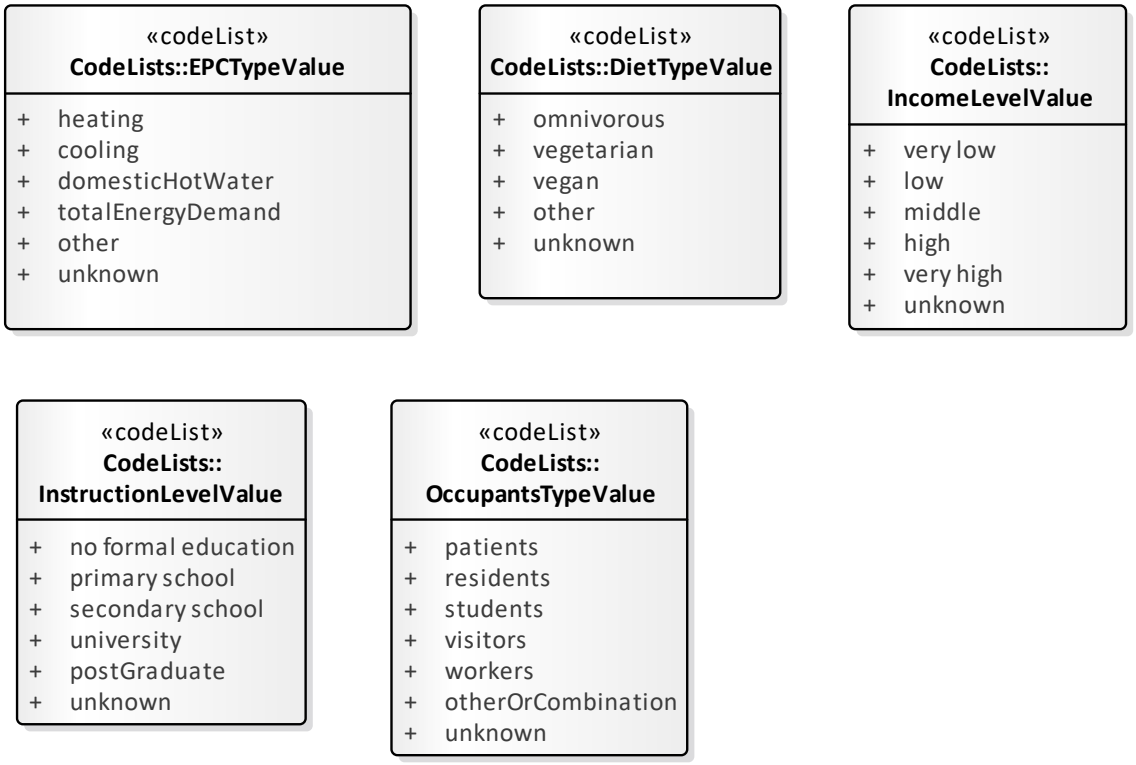
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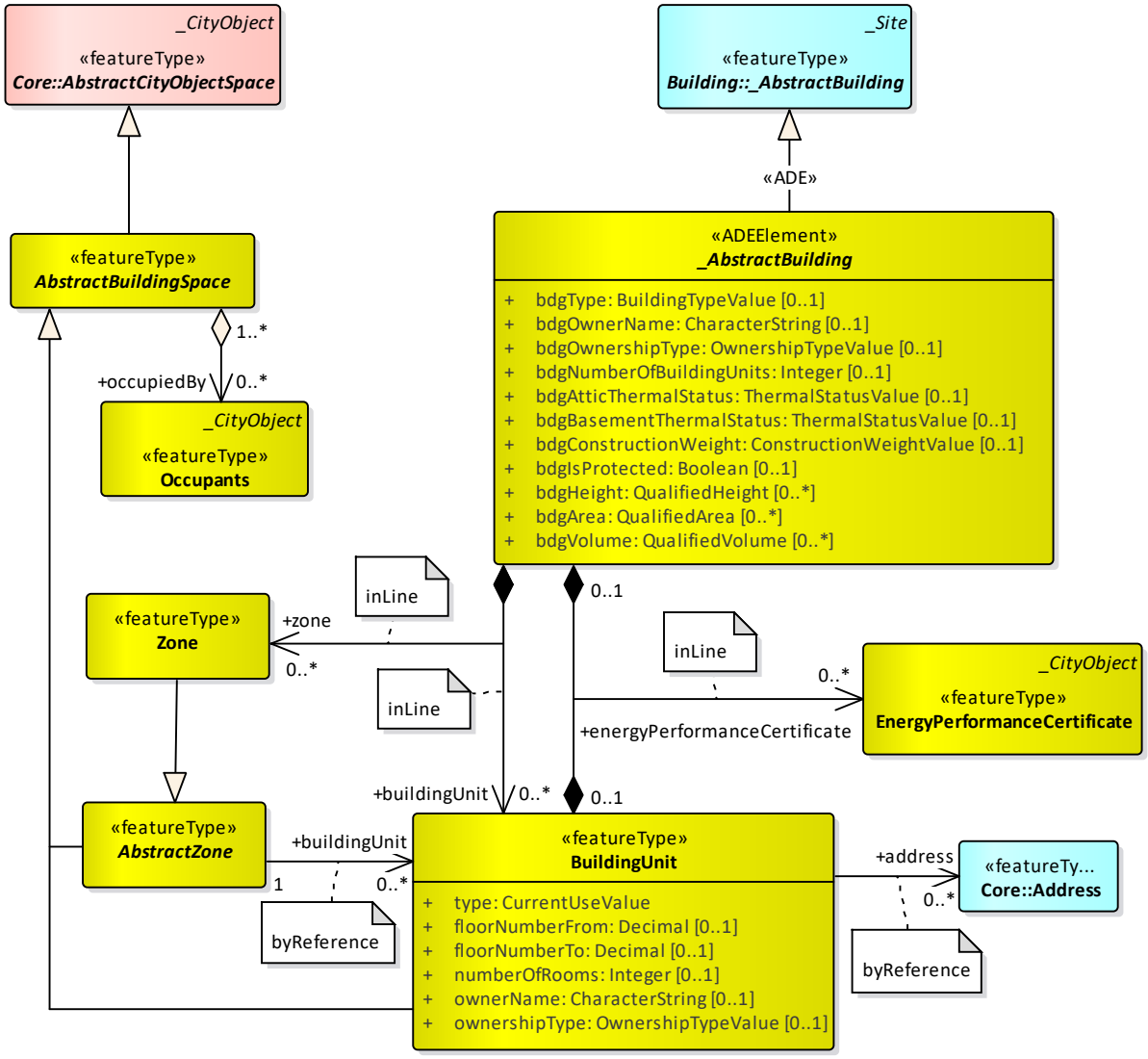
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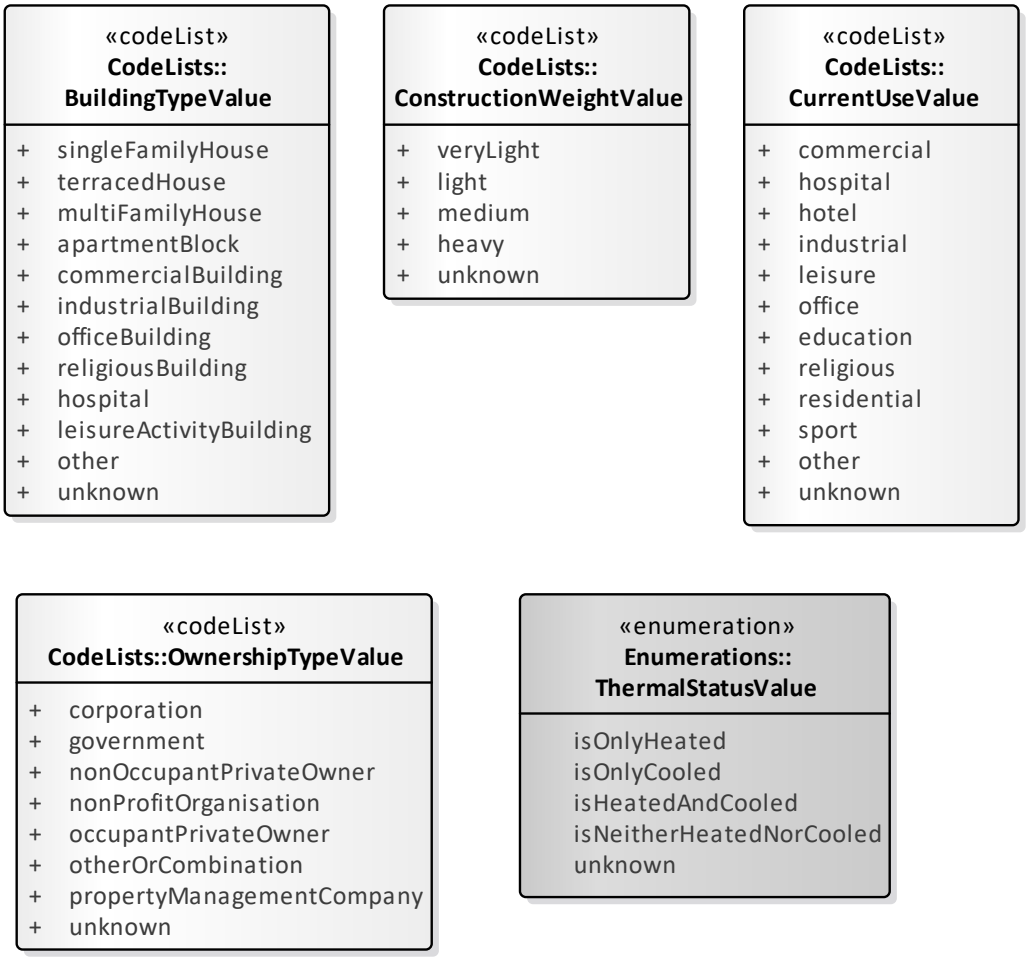
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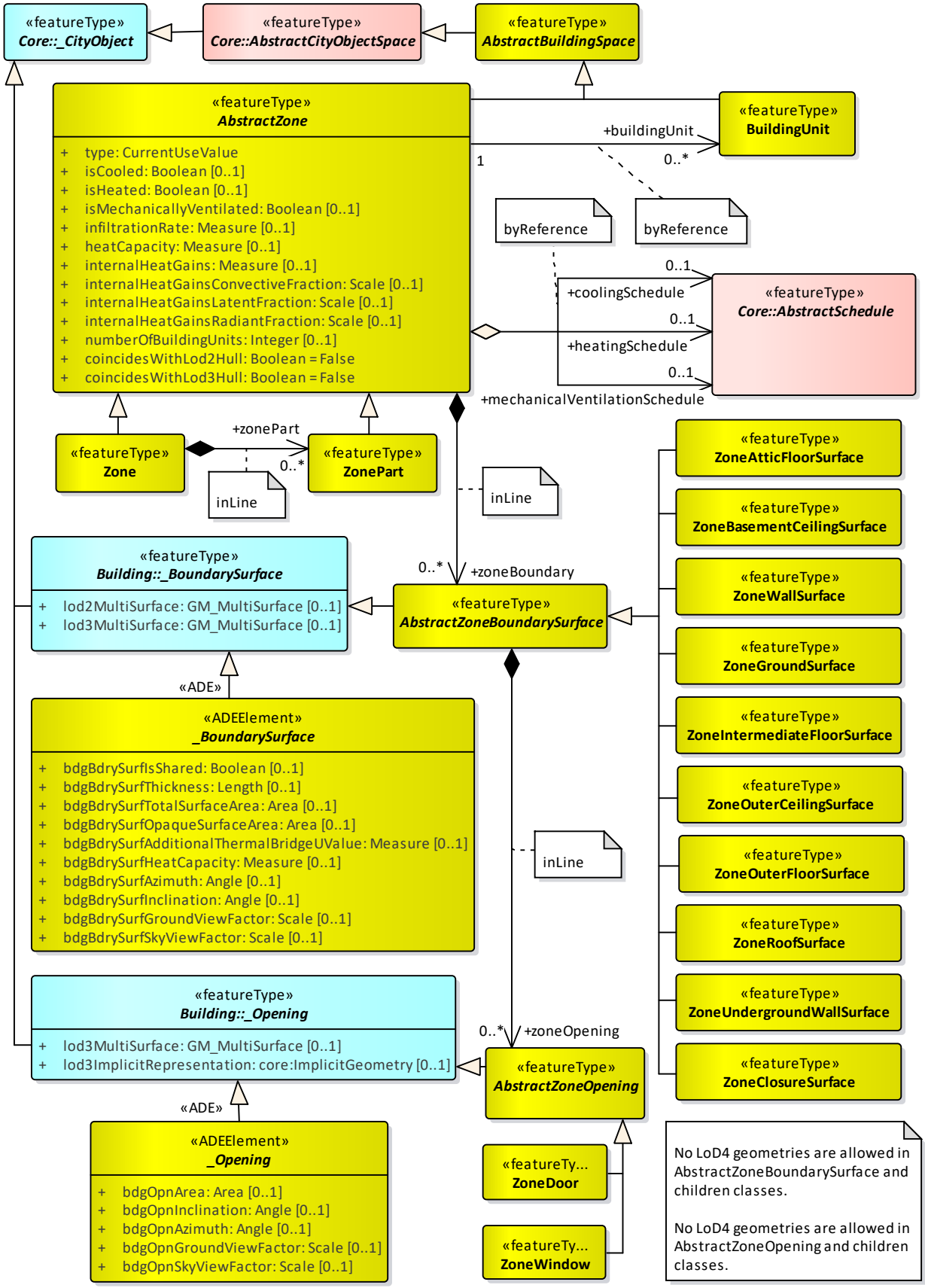
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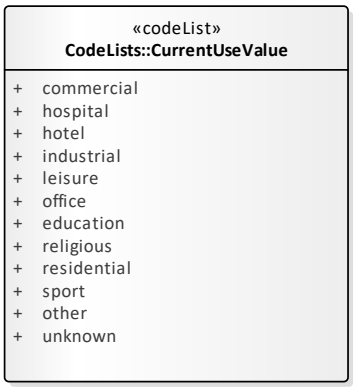
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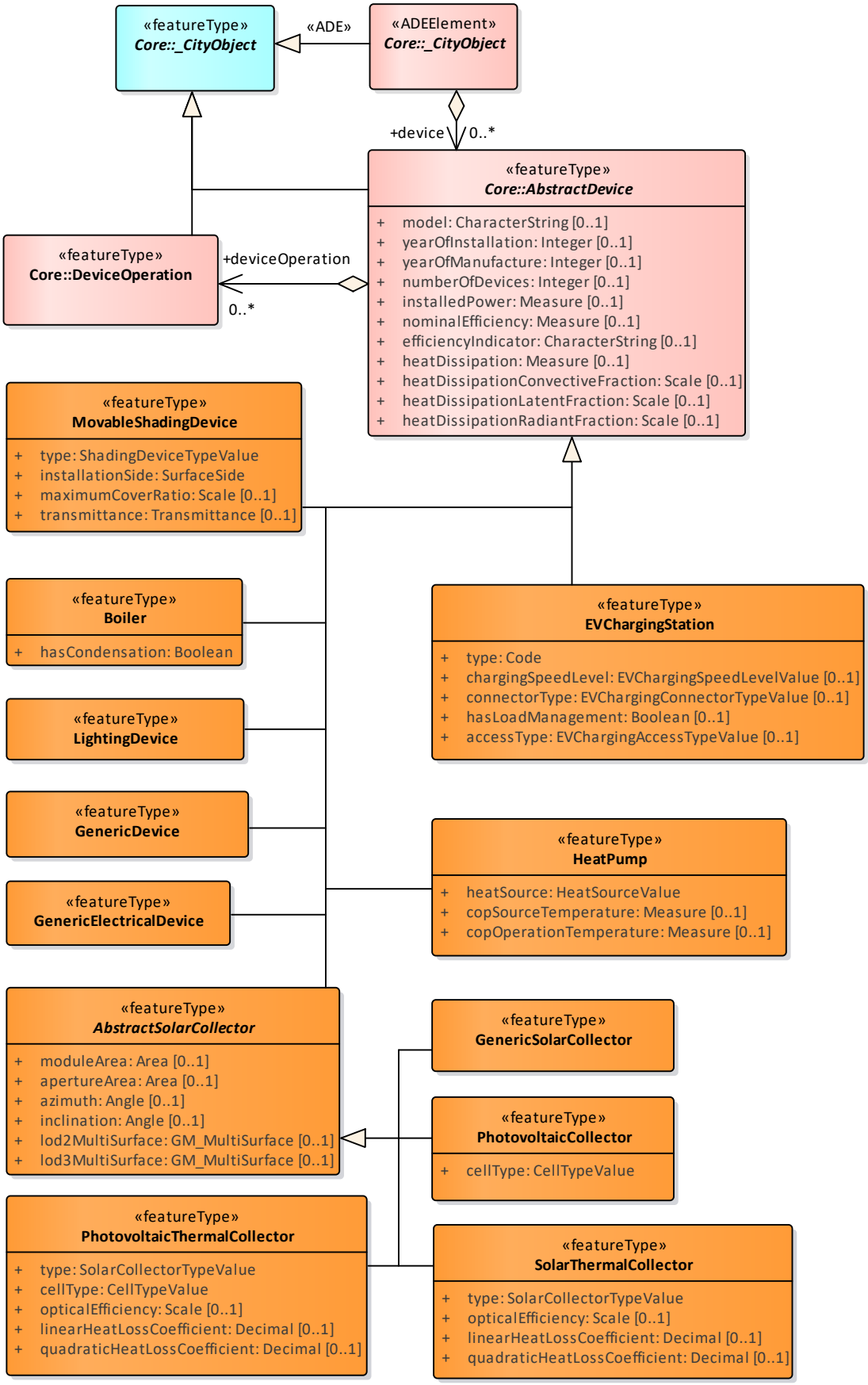
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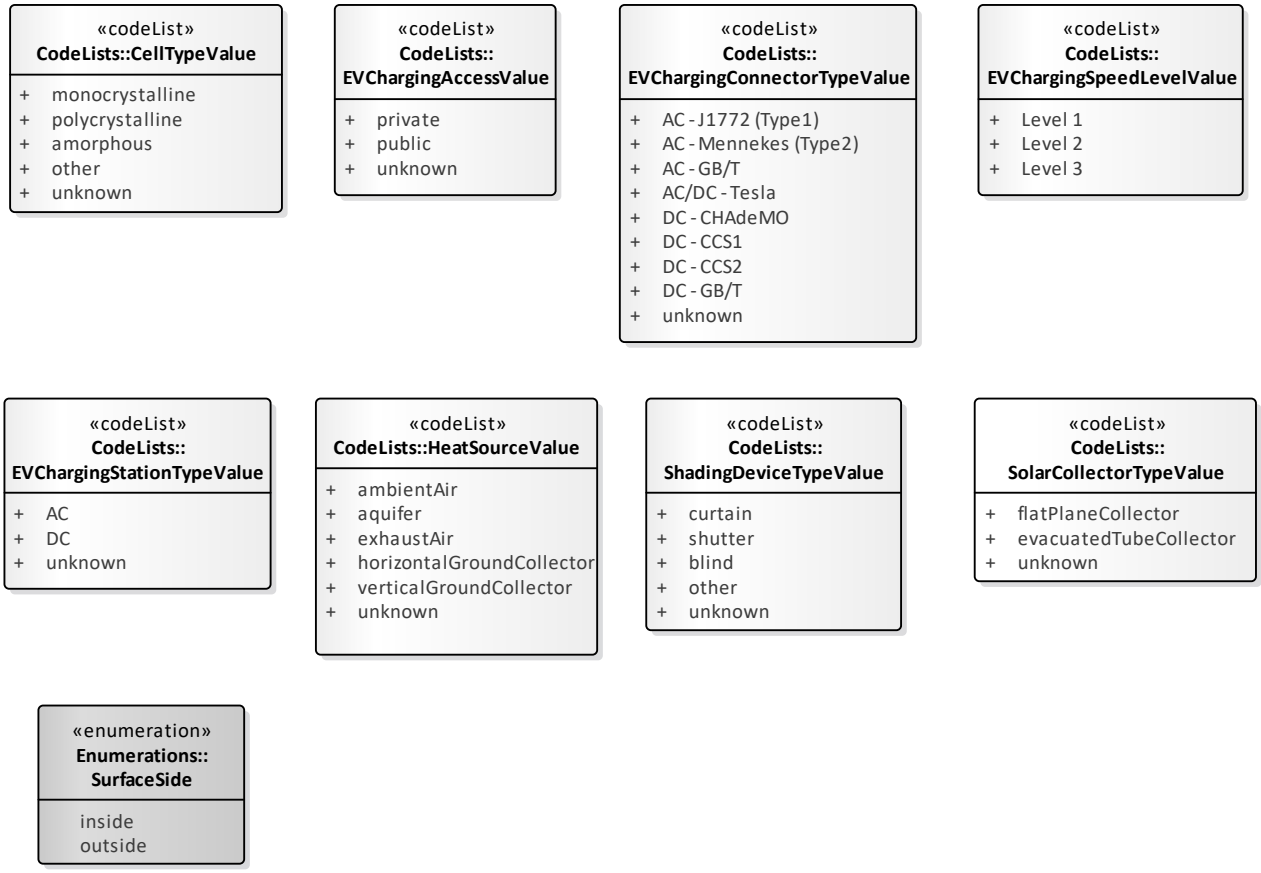
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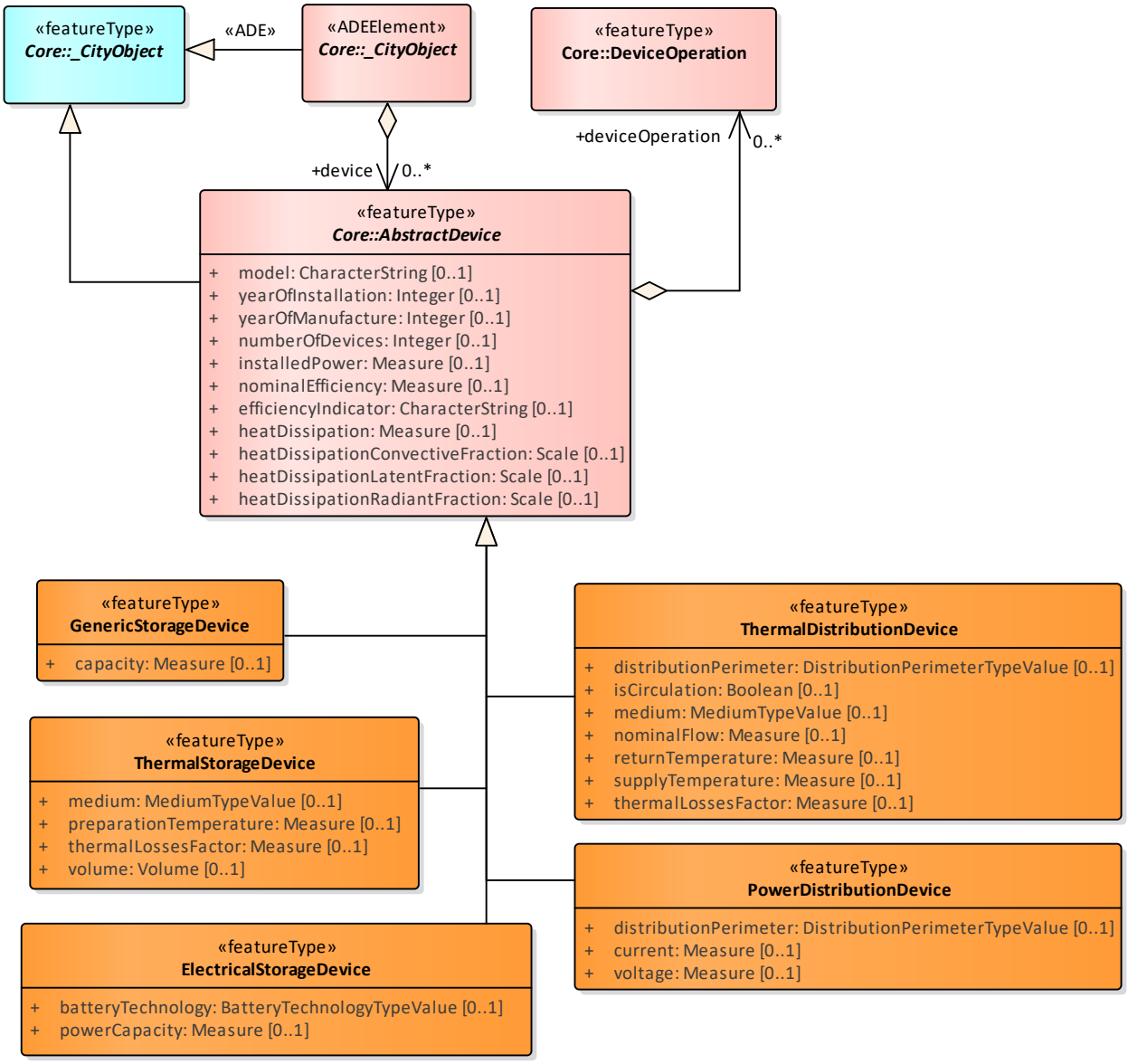
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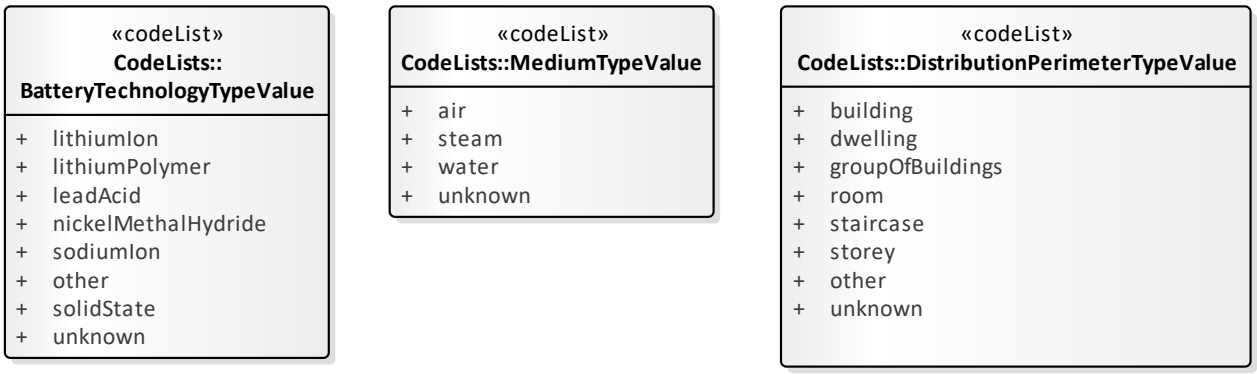
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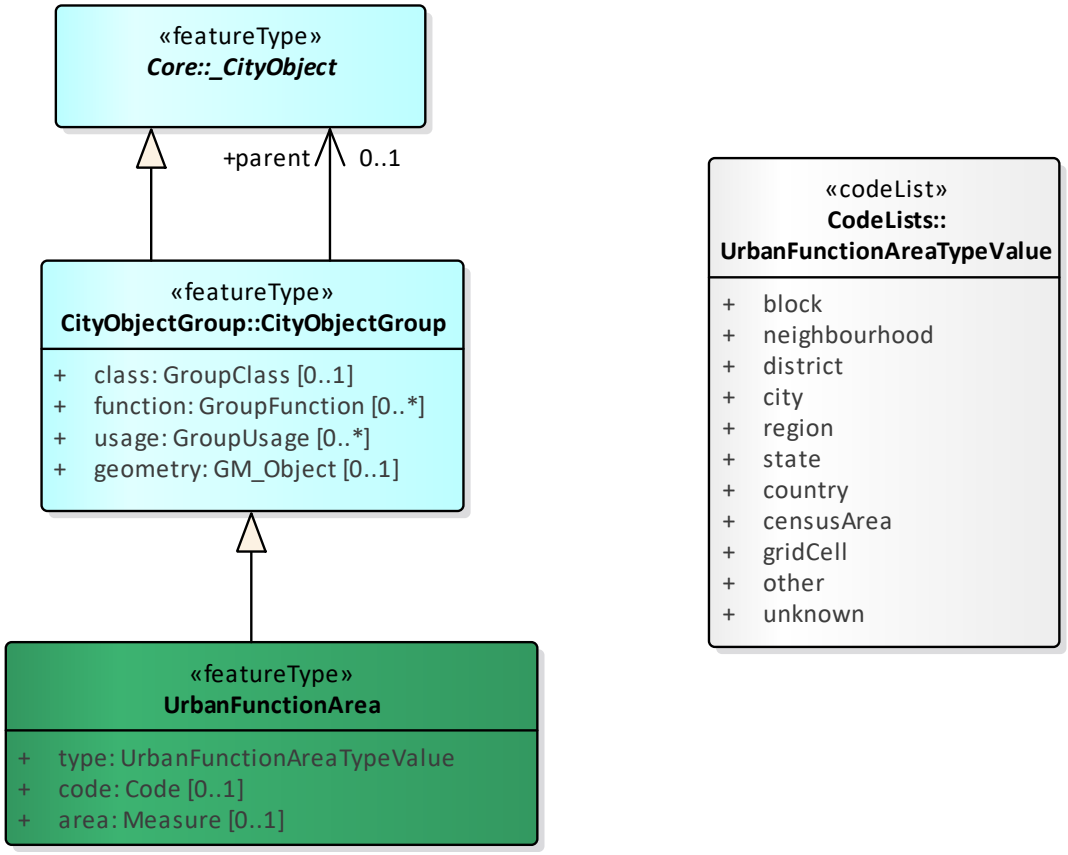
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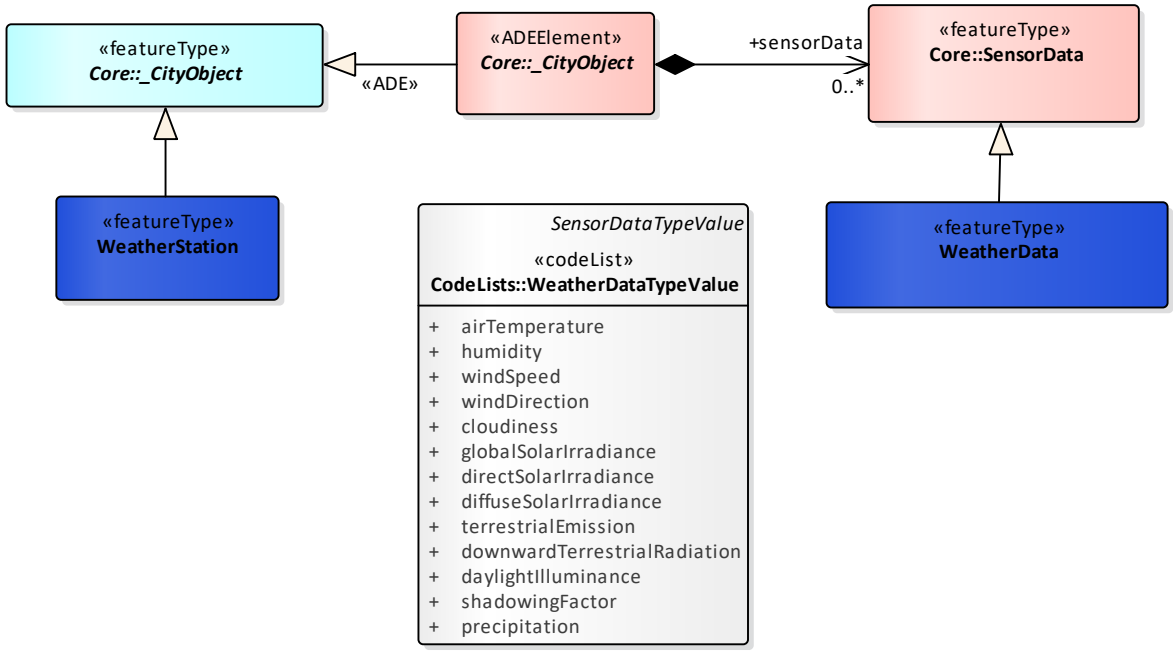
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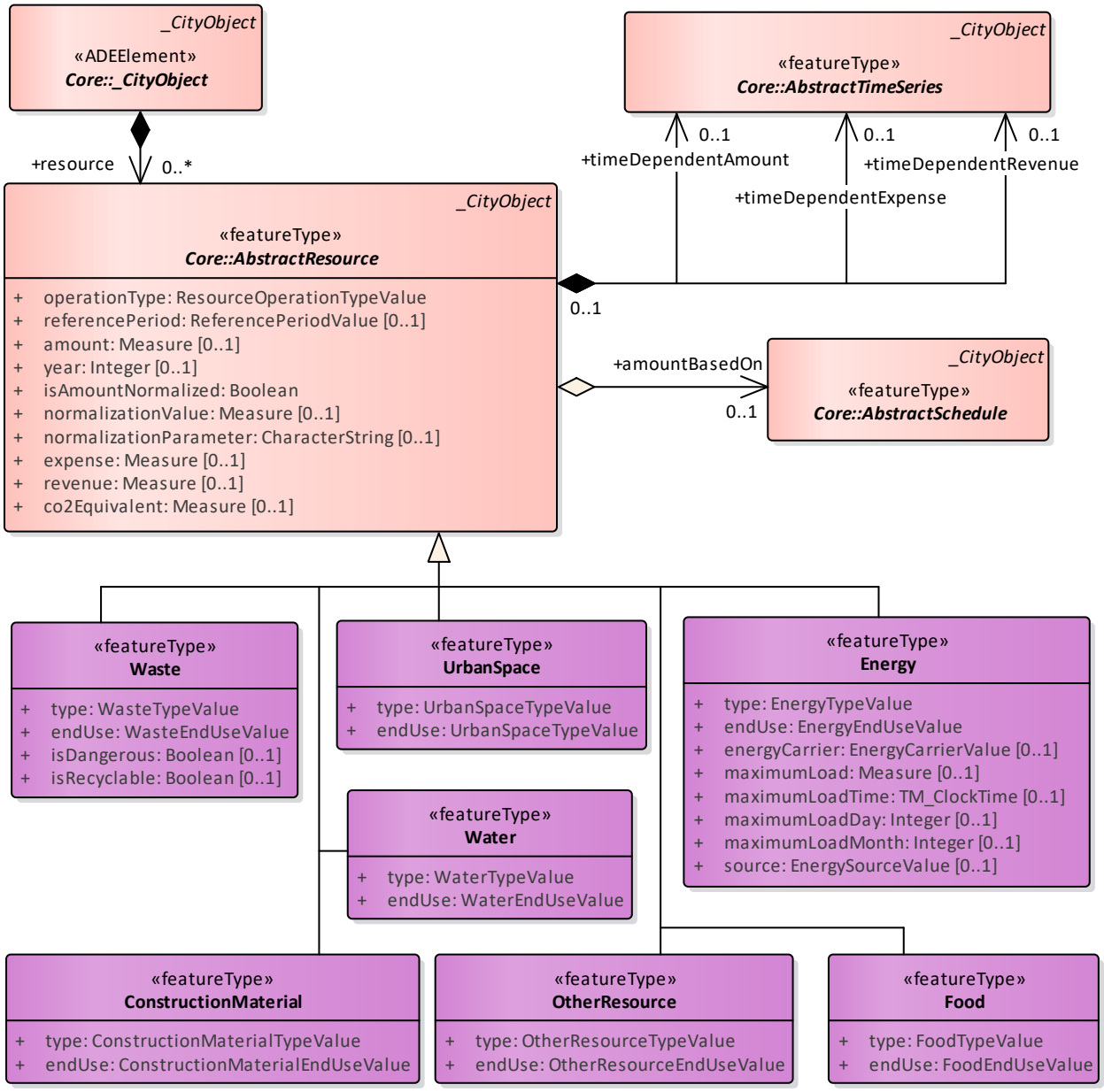
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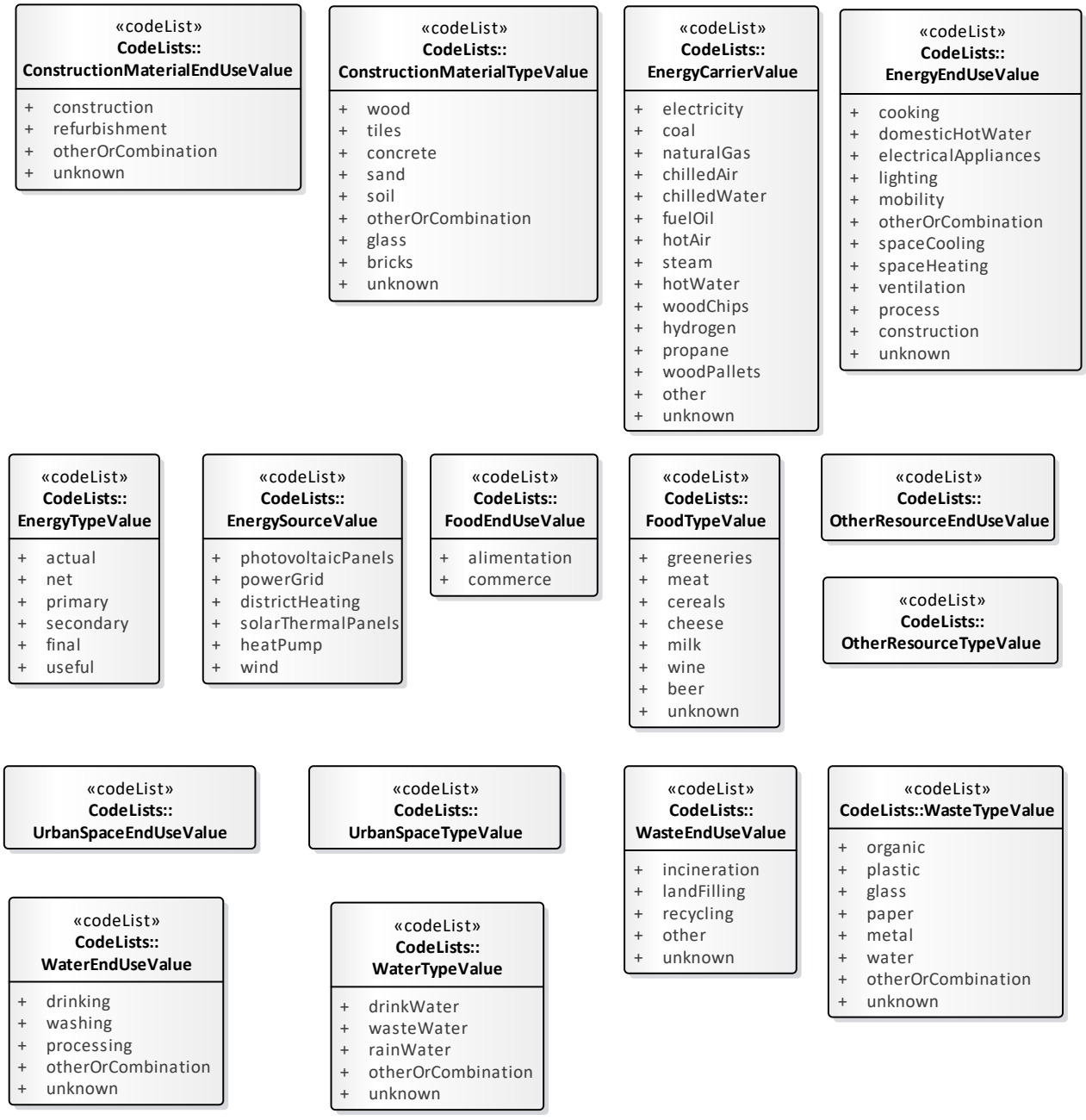
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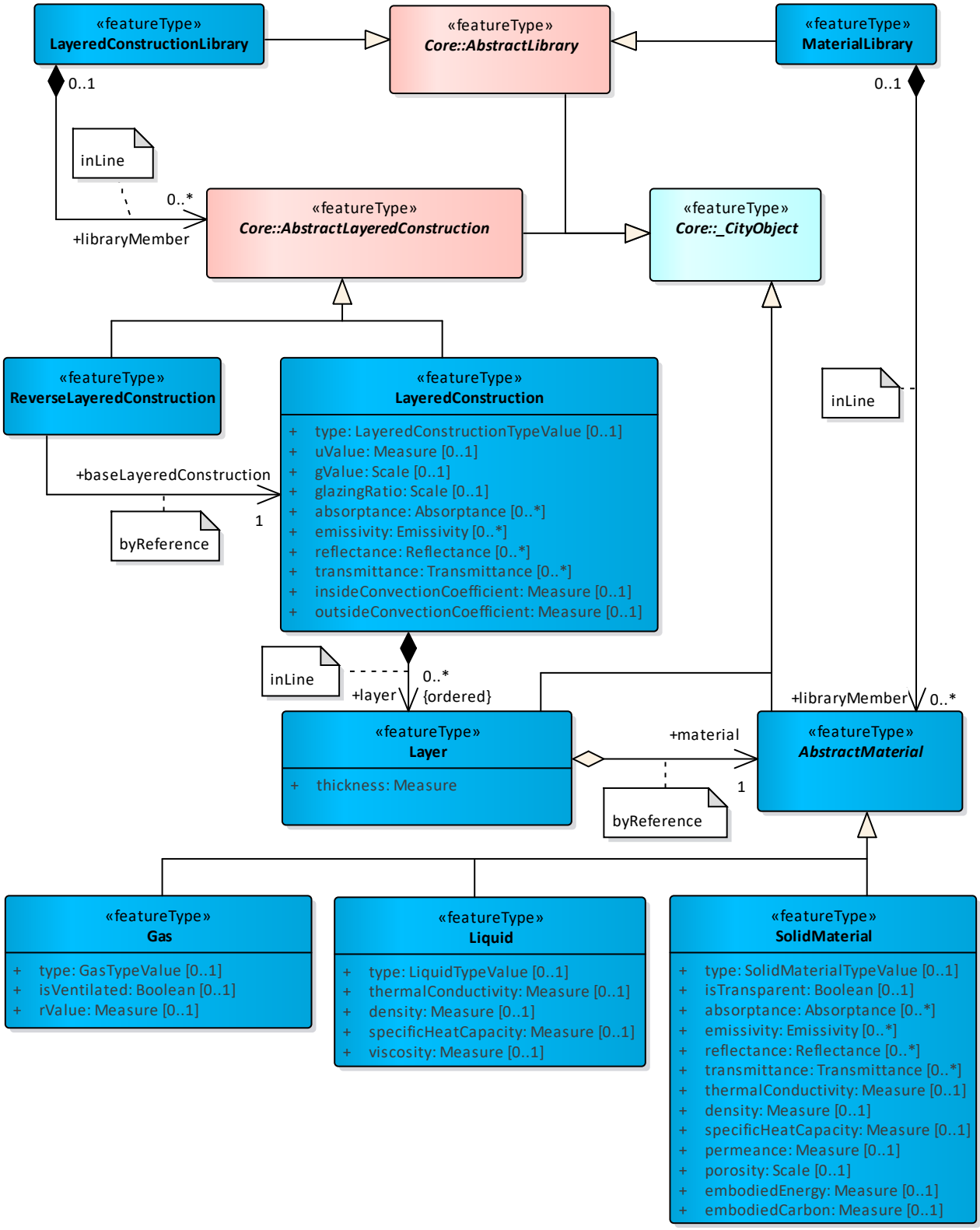
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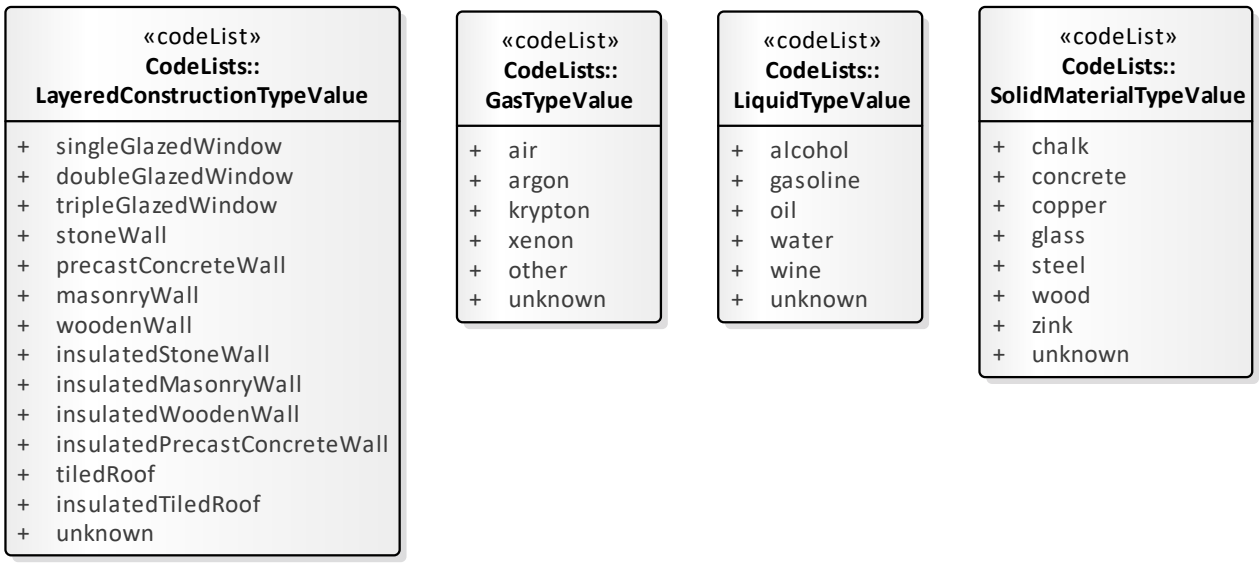
Resources package: CodeLists and Enumerations



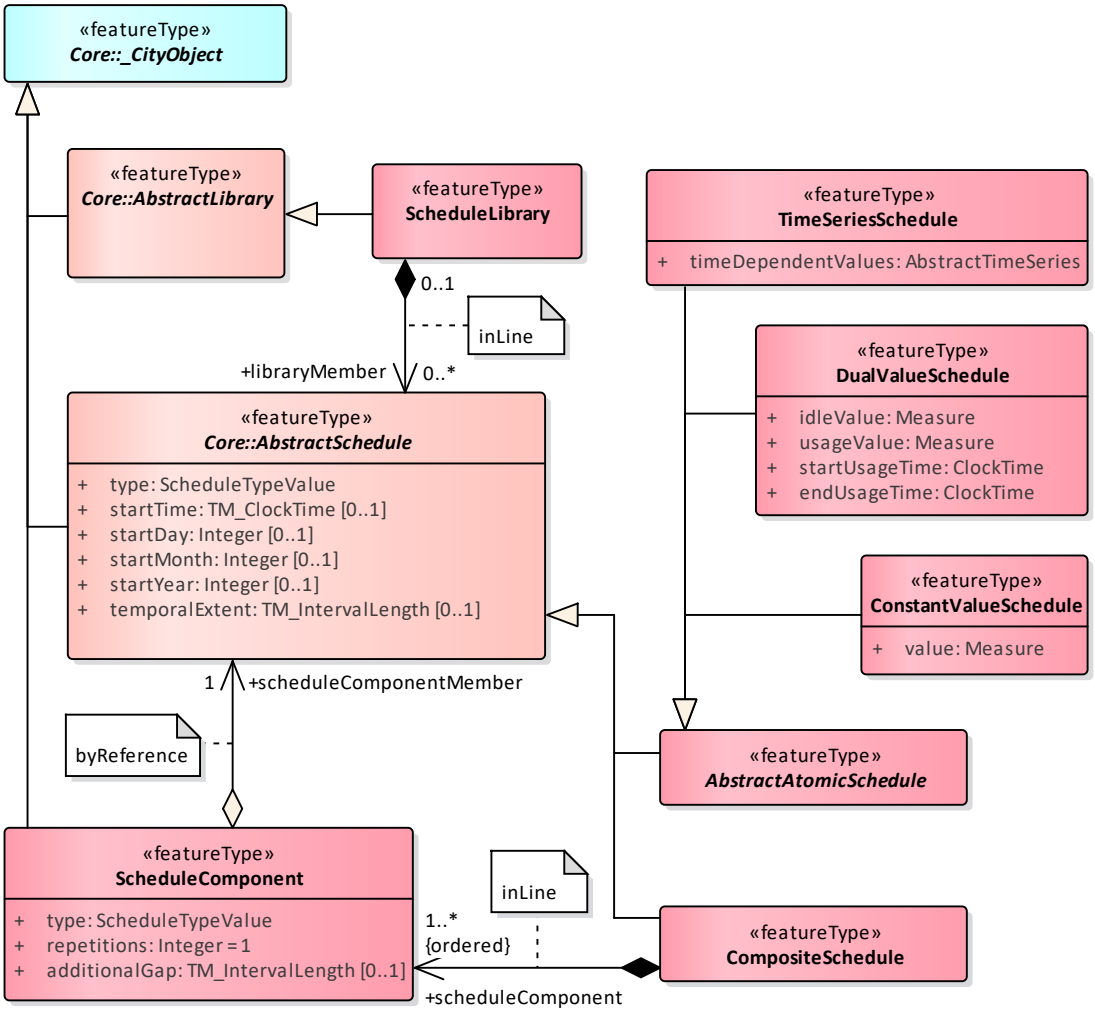
Layered construction package



Layered construction package: CodeLists



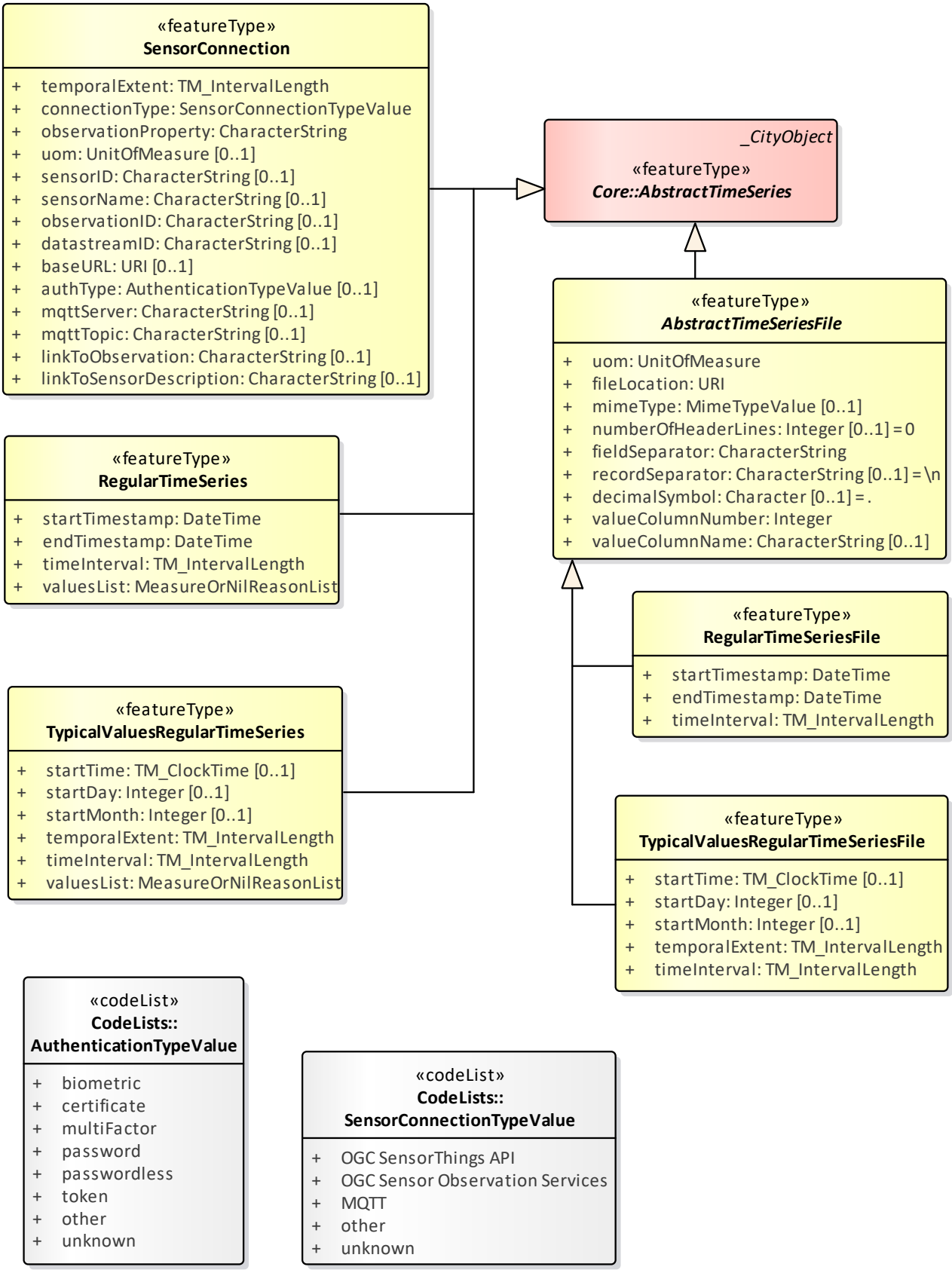
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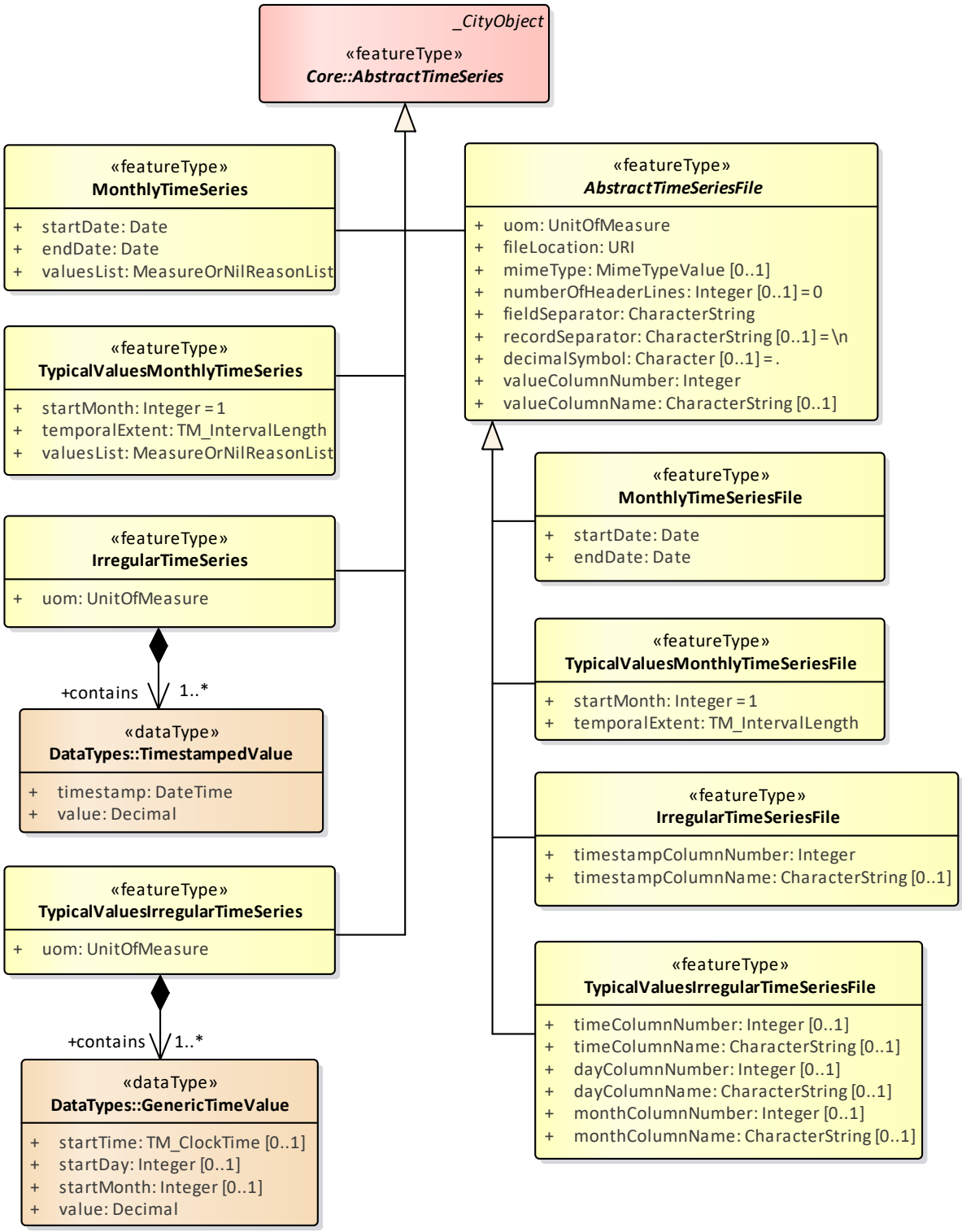
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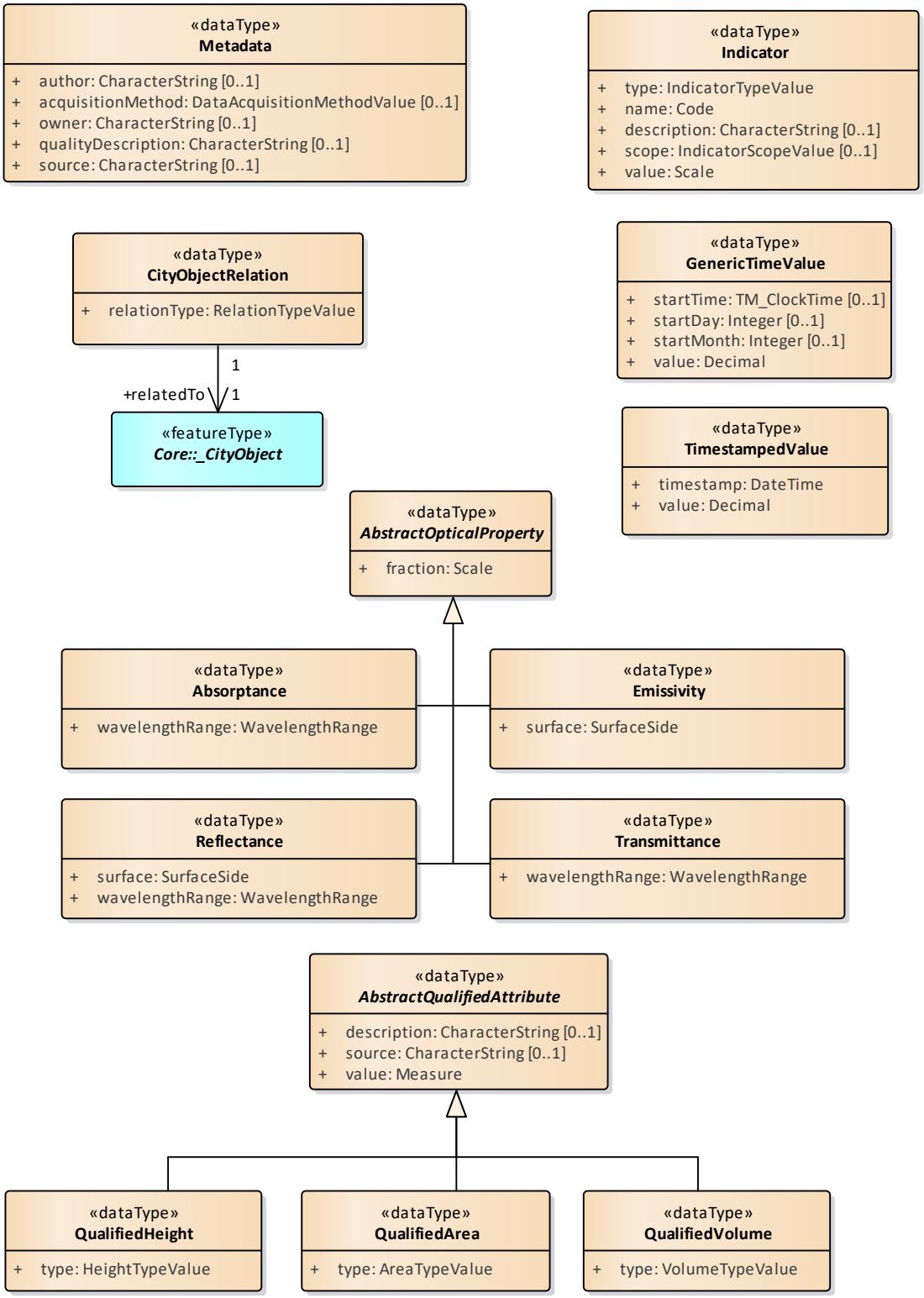
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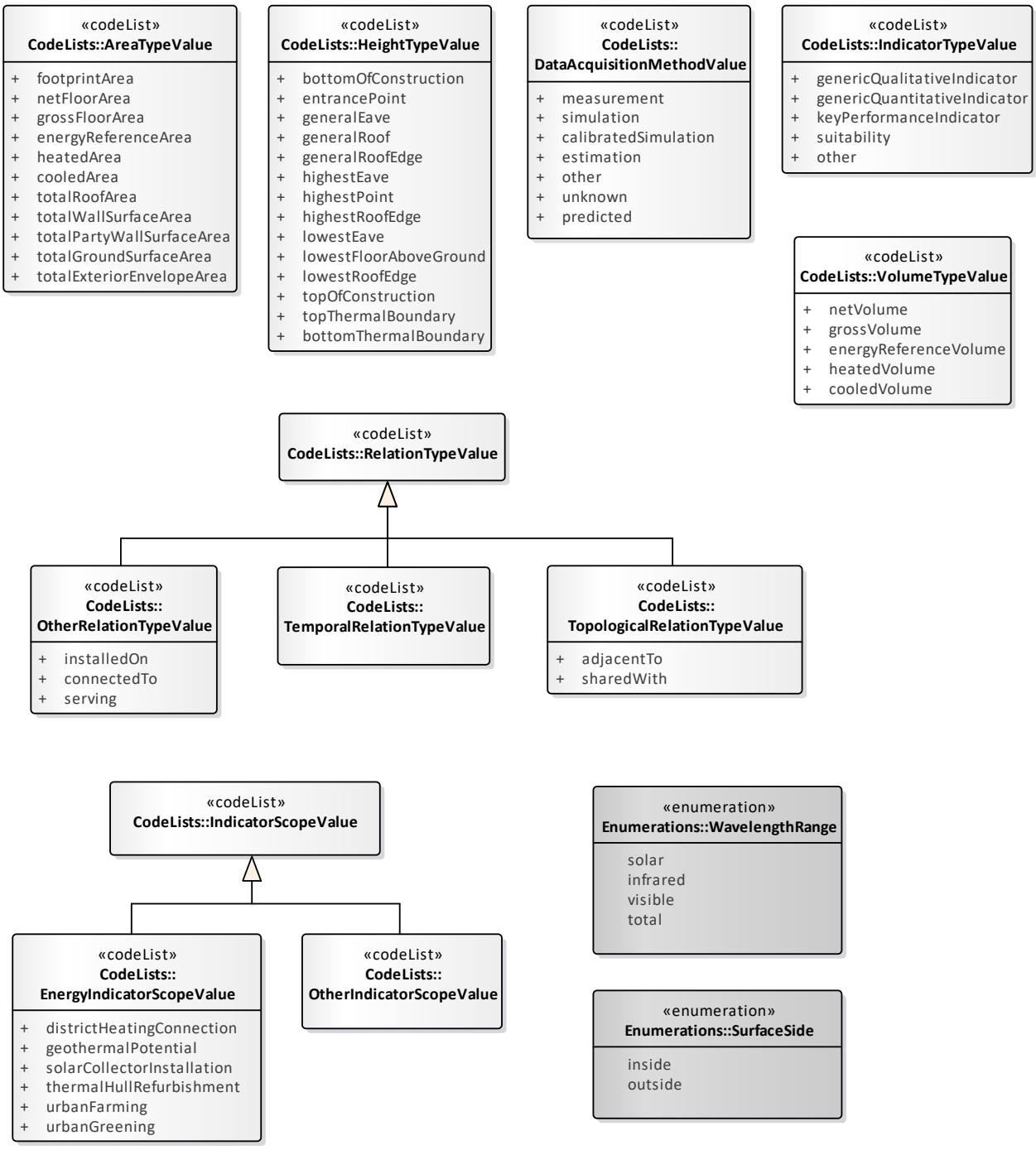
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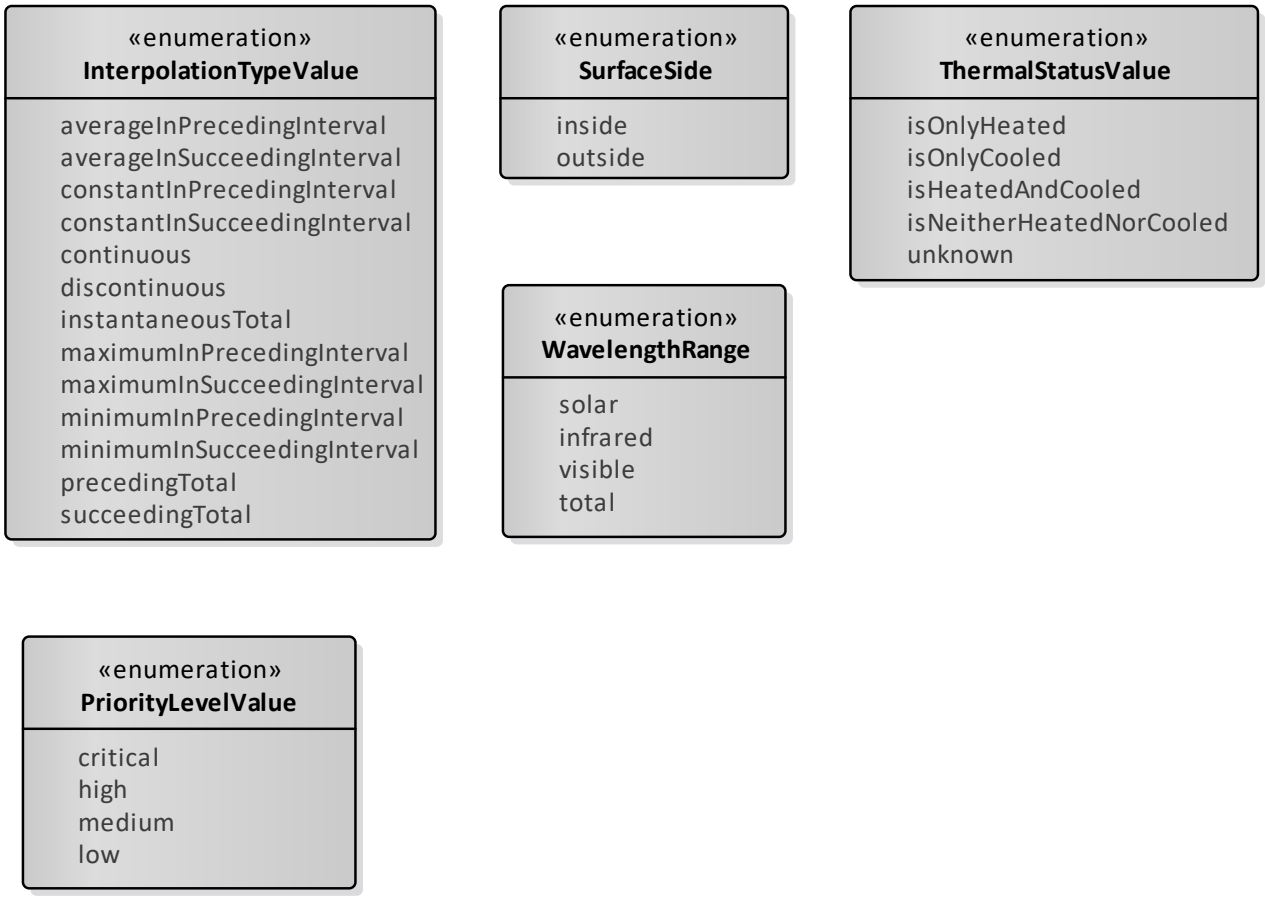
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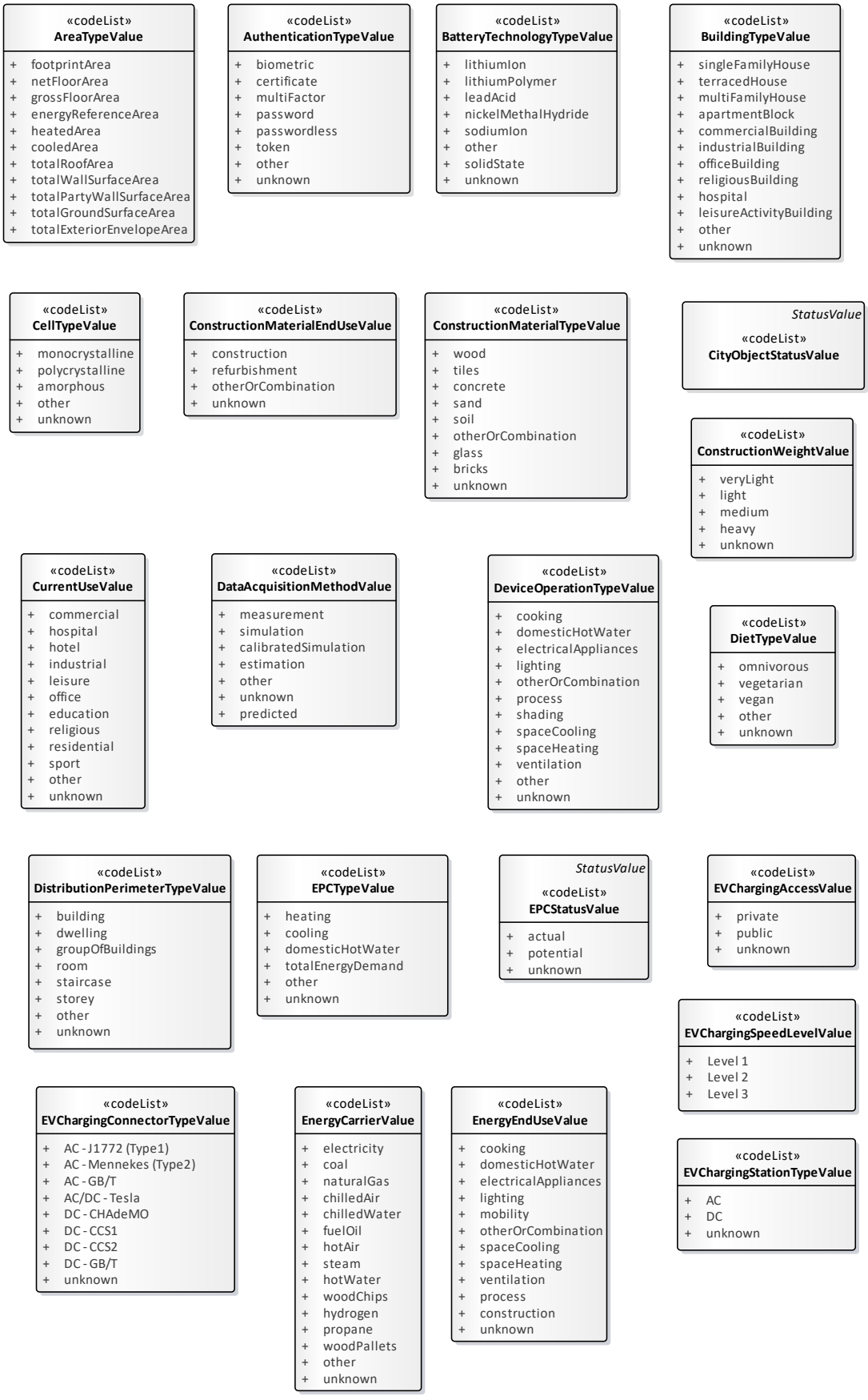
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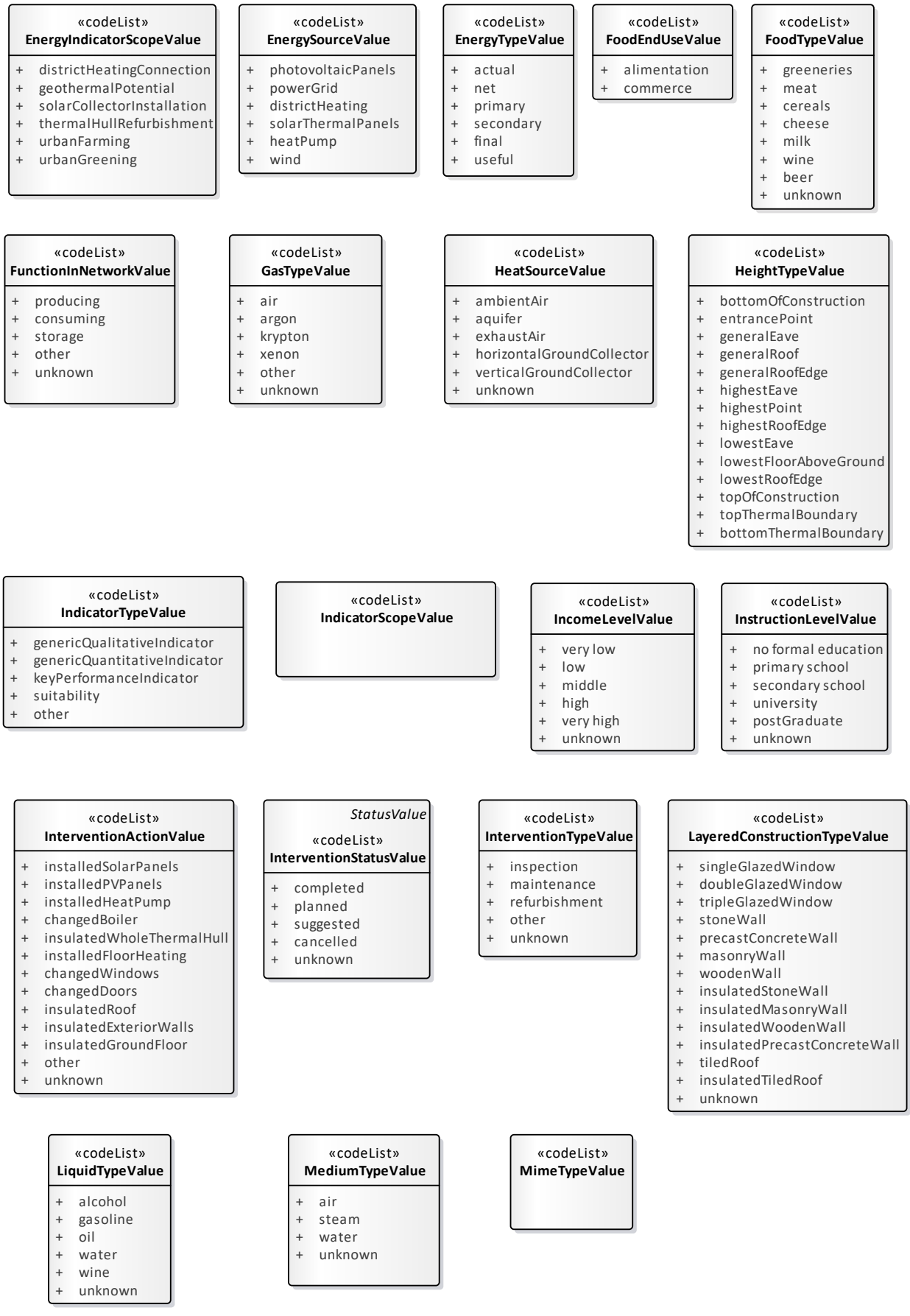


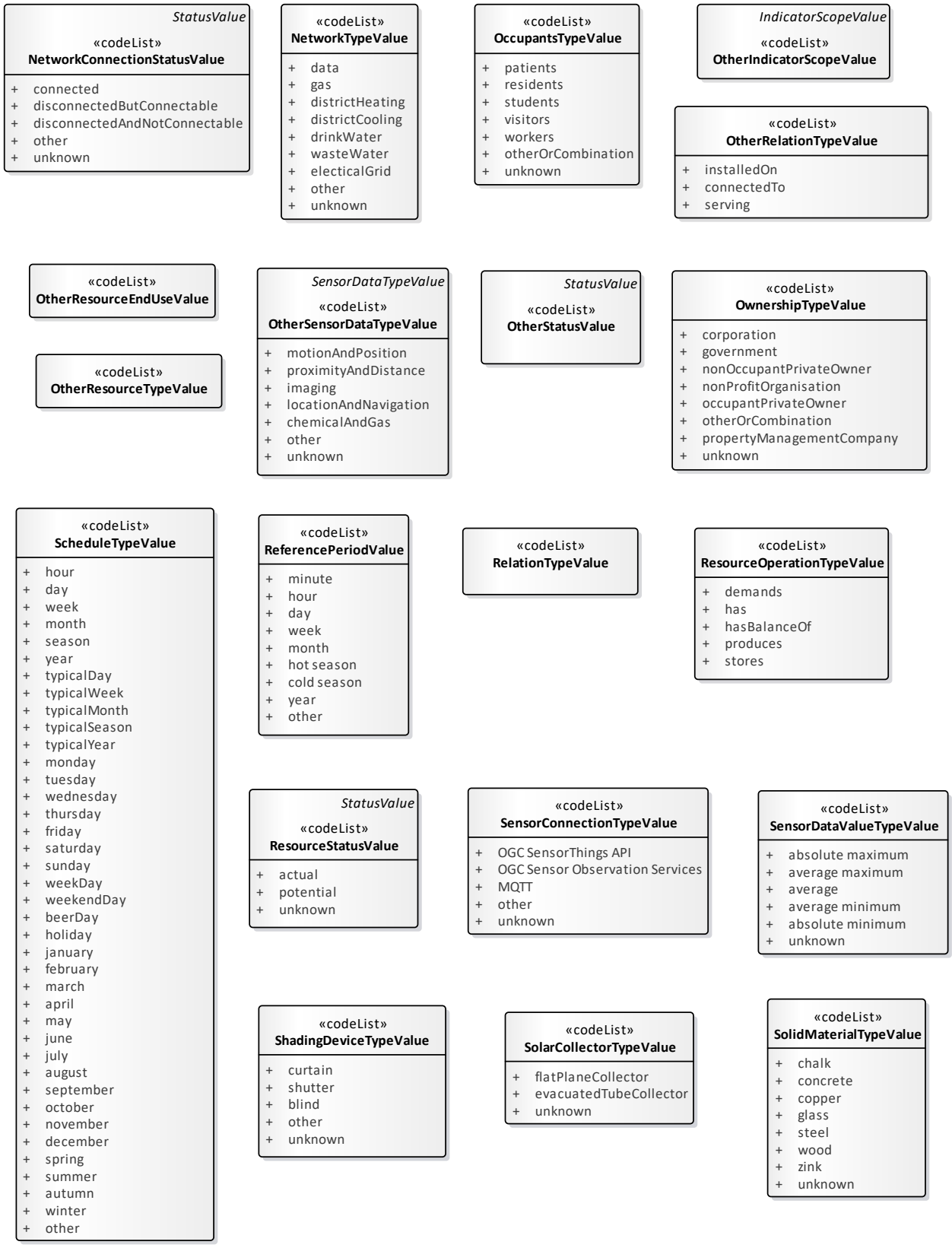
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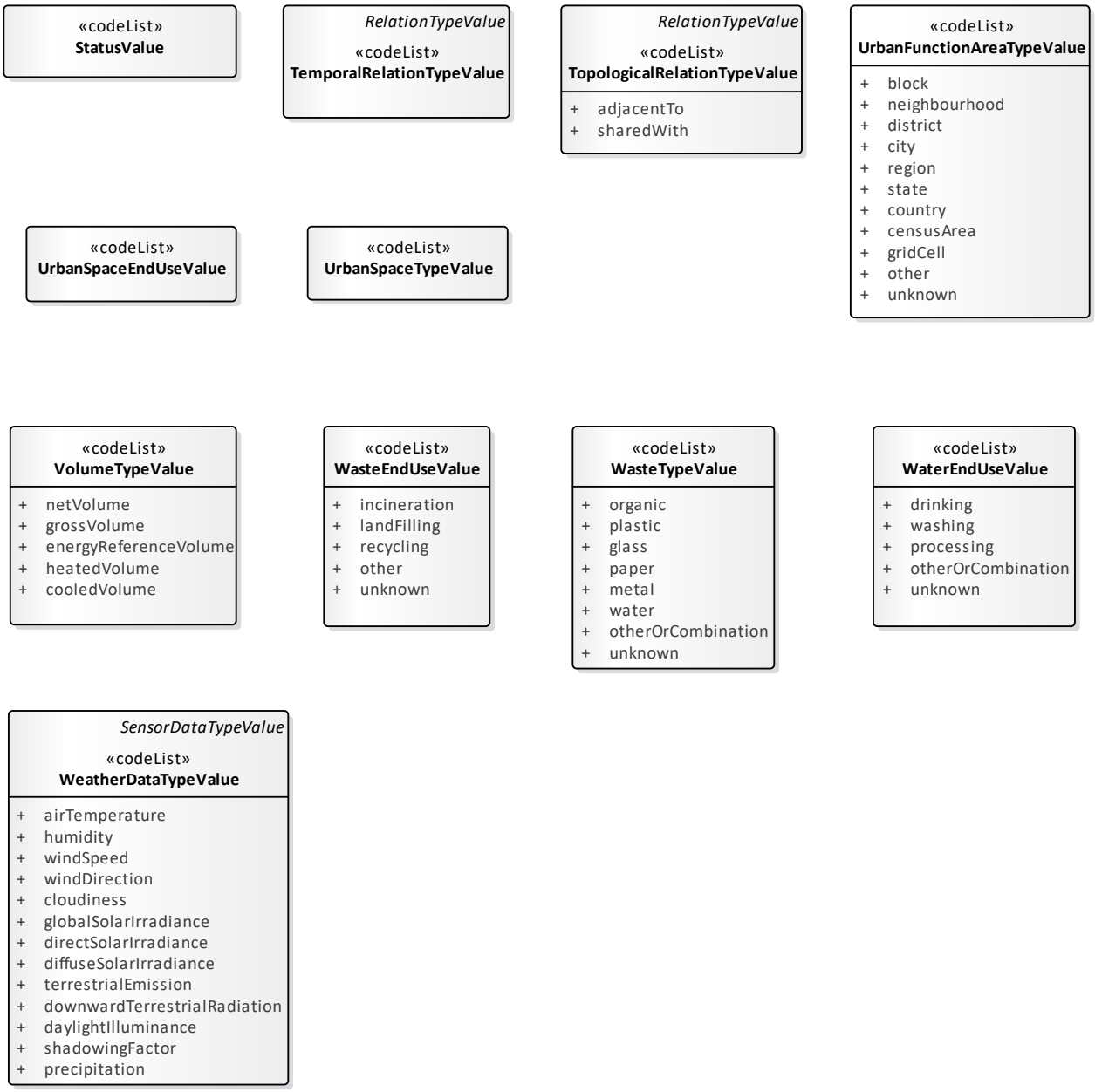


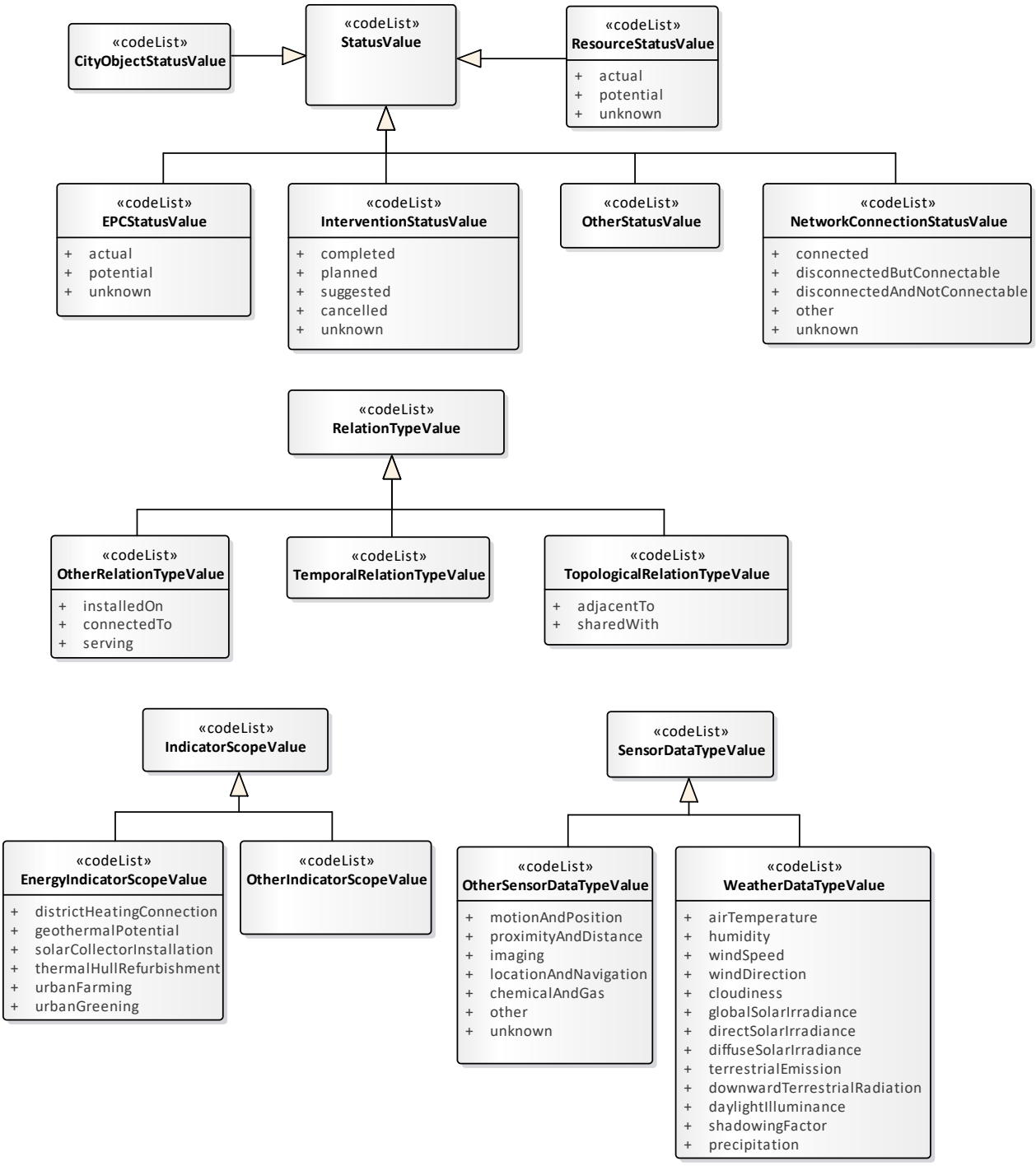
Codelists package




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+ urbanFarming
+ urbanGreening
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+ powerGrid
+ districtHeating
+ solarThermalPanels
+ heatPump
+ wind
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+ net
+ primary
+ secondary
+ final
+ useful
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FoodEndUseValue
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+ commerce
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FoodTypeValue
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+ cereals
+ cheese
+ milk
+ wine
+ beer
+ unknown
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+ consuming
+ storage
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+ krypton
+ xenon
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+ exhaustAir
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+ verticalGroundCollector
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+ lowestRoofEdge
+ topOfConstruction
+ topThermalBoundary
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+ water
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MimeTypeValue


StatusValue
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+ disconnectedAndNotConnectable
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+ districtCooling
+ drinkWater
+ wasteWater
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+ friday
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+ sunday
+ weekDay
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+ february
+ march
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+ june
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+ november
+ december
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+ summer
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+ year
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+ cooledVolume
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+ metal
+ water
+ otherOrCombination
+ unknown
«codeList»
WaterEndUseValue
+ drinking
+ washing
+ processing
+ otherOrCombination
+ unknown
SensorDataTypeValue
«codeList»
WeatherDataTypeValue
+ airTemperature
+ humidity
+ windSpeed
+ windDirection
+ cloudiness
+ globalSolarIrradiance
+ directSolarIrradiance
+ diffuseSolarIrradiance
+ terrestrialEmission
+ downwardTerrestrialRadiation
+ daylightIlluminance
+ shadowingFactor
+ precipitation


«codeList»
CityObjectStatusValue
«codeList»
StatusValue
«codeList»
ResourceStatusValue
+ actual
+ potential
+ unknown
«codeList»
EPCStatusValue
+ actual
+ potential
+ unknown
«codeList»
InterventionStatusValue
+ completed
+ planned
+ suggested
+ cancelled
+ unknown
«codeList»
OtherStatusValue
«codeList»
NetworkConnectionStatusValue
+ connected
+ disconnectedButConnectable
+ disconnectedAndNotConnectable
+ other
+ unknown
«codeList»
RelationTypeValue
«codeList»
OtherRelationTypeValue
+ installedOn
+ connectedTo
+ serving
«codeList»
TemporalRelationTypeValue
«codeList»
TopologicalRelationTypeValue
+ adjacentTo
+ sharedWith
«codeList»
IndicatorScopeValue
«codeList»
EnergyIndicatorScopeValue
+ districtHeatingConnection
+ geothermalPotential
+ solarCollectorInstallation
+ thermalHullRefurbishment
+ urbanFarming
+ urbanGreening
«codeList»
OtherIndicatorScopeValue
«codeList»
SensorDataTypeValue
«codeList»
OtherSensorDataTypeValue
+ motionAndPosition
+ proximityAndDistance
+ imaging
+ locationAndNavigation
+ chemicalAndGas
+ other
+ unknown
«codeList»
WeatherDataTypeValue
+ airTemperature
+ humidity
+ windSpeed
+ windDirection
+ cloudiness
+ globalSolarIrradiance
+ directSolarIrradiance
+ diffuseSolarIrradiance
+ terrestrialEmission
+ downwardTerrestrialRadiation
+ daylightIlluminance
+ shadowingFactor
+ precipitation

Changelog

Version 3.0 beta 8

2026-10-16

Enumerations package:
- Added class **PriorityLevelValue**

Core package:
- Class **Intervention:**
  - Added property *priorityLevel*
  - Added property *plannedStartDate*
  - Added property *plannedEndDate*
  - Renamed property *interventionStartDate* to *actualStartDate*
  - Renamed property *interventionEndDate* to *actualEndDate*

Time series package:
- Class **RegularTimeSeries:**
  - Changed type of property *valuesList* to *gml:MeasureOrNullListType*
- Class **TypicalValuesRegularTimeSeries:**
  - Changed type of property *valuesList* to *gml:MeasureOrNullListType*
- Class **MonthlyTimeSeries:**
  - Changed type of property *valuesList* to *gml:MeasureOrNullListType*
- Class **TypicalValuesMonthlyTimeSeries:**
  - Changed type of property *valuesList* to *gml:MeasureOrNullListType*

2026-09-20

Layered Construction package
- Class **Gas**
  - Dropped default value of property *isVentilated*

Time series package:
- Class **TypicalValuesMonthlyTimeSeries:**
  - Changed cardinality of property *startMonth* to 1
- Class **TypicalValuesMonthlyTimeSeriesFile:**
  - Changed cardinality of property *startMonth* to 1

2026-08-25

Building package:
- Class **ADE _AbstractBuilding**
  - - Changed target of property *zone* from **AbstractZone** to **Zone**

Layered Construction package
- Class **SolidMateria**
  - Changed multiplicity of property *isTransparent* to 0..1

2026-08-08

Core package:
- Class **SensorData:**
  - Dropped property *position*

Devices package:
- Renamed class **PowerDistrubution** to **PowerDistributionDevice**

- Renamed class **ThermalDistrubution** to **ThermalDistributionDevice**

Layered Construction package
- Class **MaterialLibrary**
  - Changed multiplicity of property *libraryMember* to 0..*
- Class **LayeredConstructionLibrary**
  - Changed multiplicity of property *libraryMember* to 0..*

Schedules package:
- Class **ScheduleLibrary**
  - Changed multiplicity of property *libraryMember* to 0..*

Cumulative list of changes since beta 7 (as of 2026-05-14)

CodeLists package:
Several CodeList classes added and some renamed. Added values to several CodeLists.

Enumerations package:
- Class **NetworkConnectionStatusValue** has been changed to a CodeList and moved to the corresponding package.

DataTypes package:
- Dropped class **Suitability**
- Added class **Absorptance**
- Added class **CityObjectRelation**
- Added class **FeatureRelation**
- Added class **GenericTimeValue**
- Added class **Indicator**
- Added class **Metadata**
- Added class **TimestampedValue**

Core package:
- Dropped class **AbstractBuildingUnit**
- Dropped class **AbstractUsageZone**
- Dropped class **AbstractThermalZone**
- Dropped class **RefurbishmentMeasure**
- Moved class **EnergyPerformanceCertification** to Building package
- Moved class **ADE _AbstractBuilding** to Building package
- Renamed class **AbstractBuildingPartition** to **AbstractCityObjectSpace**
- Renamed class **WeatherData** to **SensorData**
- Added class **AbstractADEFeature**
- Added class **AbstractFeatureWithLifeSpan**
- Added class **Intervention**
- Classes **AbstractResource**, **AbstractSchedule**, **AbstractLayeredConstruction**, **UtilityNetworkCnnection**, **SensorData**, **DeviceOperation** are now derived from class **AbstractFeatureWithLifeSpan**
- Class **ADE _CityObject:**
  - Dropped property *suitability*
  - Renamed property from *weatherData* to *sensorData*
  - Added property *metadata*
  - Added property *identifier*
  - Added property *validFrom*
  - Added property *validTo*
  - Added property *status*
  - Added property *indicator*
- Class **AbstractLibrary:**
  - Dropped property *source*
  - Dropped property *author*
- Class **AbstractCityObjectSpace:**
  - Added property *lod0MultiSurface*

- Class **AbstractSchedule:**
  - Dropped property *libraryCode*
- Class **AbstractResource:**
  - Dropped property *status*
  - Dropped property *amountType*
  - Renamed property *costsMoney* to *expense*
  - Renamed property *yieldsMoney* to *revenue*
  - Added property *referencePerdiod*
  - Added property *amoundBasedOn*
  - Added property *timeDependentExpense*
  - Added property *timeDependentRevenue*
- Class **AbstractTimeSeries:**
  - Dropped property *acquisitionMethod*
  - Dropped property *source*
- Class **UtilityNetworkConnection:**
  - Dropped property *connectionStatus*

Building physics package:
Dropped. Some classes reworked and moved to new module Building

Occupancy package:
Dropped. Some classes reworked and moved to new module Building

Building package:
This module is new. It reworks and merges the previous Building physics module and Occupancy module. Additionally, it incorporates also class *ADE _AbstractBuilding* and its subclasses.

- Class **ADE _AbstractBuilding**
  - Class is moved from Core module
  - Dropped property *usageZone*
  - Dropped property *thermalZone*
  - Added property *occupiedBy*
  - Added property *zone*
  - Added property *owner*
  - Added property *ownershipType*
  - Added property *numberOfBuildingUnits*
- Class **EnergyPerformanceCertificate**
  - Class is moved from Core module
  - Class is derived from class **AbstractADEFeatureWithLifeSpan**
  - Dropped property *type*
  - Dropped property *issueDate*
  - Dropped property *expirationDate*
  - Dropped property *certificationUri*
- Class **Occupants**
  - Class is taken over from former Occupancy module
  - Class is now derived from class **AbstractADEFeatureWithLifeSpan**
  - Renamed relation *occupancyRate* to *occupancySchedule*
- Added class **AbstractBuildingSpace**
- Class **BuildingUnit**
  - Class is taken over from former Occupancy module
  - Class is now derived from class **AbstractBuildingSpace**
  - Added property *floorNumberFrom*
  - Added property *floorNumerTo*
- Added class **AbstractZone**
- Added class **Zone**
- Added class **ZonePart**
- Class **ADE _BoundarySurface**
  - Class is taken over from former Building physics module
  - Dropped property *bdgBdrySurfIsAdiabatic*

  - Added property *bdgBdrySurfIsShared*
  - Added property *bdgBdryAdditionalThermalBridgeUValue*
- Added class **AbstractZoneBoundarySurface**
- Added class **ZoneAtticFloorSurface**
- Added class **ZoneBasementCeilingSurface**
- Added class **ZoneExteriorWallSurface**
- Added class **ZoneGroundSurfaceSurface**
- Added class **ZoneIntermediateFloorSurface**
- Added class **ZoneRoofSurface**
- Added class **ZoneUndergrounSurface**
- Added class **ZoneClosureSurface**
- Added class **ADE _Opening**
  - Class is taken over from former Building physics module
- Added class **ADE AbstractZoneOpening**
- Added class **ZoneDoor**
- Added class **ZoneWindow**

Devices package:
- Added class **GenericStorageDevice**
- Added class **EVChargingStation**
- Added class **ThermalDistribution**
- Added class **PowerDistribution**
- Class **AbstractDevice**
  - Added property *yearOfInstallation*

Urban function area package
- Class **UrbanFunctionArea**
  - Added property *area*
  - Set type of property *code* to gml:Code

WeatherStation package
- Added class **WeatherData**

Resources package
- Class **Energy**
  - Added property *maximulLoad*
  - Added property *maximulLoadTime*
  - Added property *maximulLoadDay*
  - Added property *maximulLoadMonth*

Layered Construction package
- Class **MaterialLibrary**
  - Changed multiplicity of property *libraryMember* to 1..*
- Class **LayeredConstructionLibrary**
  - Changed multiplicity of property *libraryMember* to 1..*
- Class **LayeredConstructionMaterial**
  - Dropped property *libraryCode*
  - Added property *type*
  - Added property *absorptance*
  - Added property *insideConvectionCoefficient*
  - Added property *outsideConvectionCoefficient*
- Class **AbstractMaterial**
  - Class is now derived from class **AbstractADEFeatureWithLifeSpan**
  - Dropped property *libraryCode*
- Class **Layer**
  - Class is now derived from class **AbstractADEFeatureWithLifeSpan**
- Class **SolidMaterial**
  - Added property *type*
  - Added property *isTransparent*

  - Added property *absorptance*
  - Added property *emissivity*
  - Added property *reflectance*
  - Added property *transmittance*
- Added class **Liquid**
- Class **Gas**
  - Added property *type*

Schedules package:

- Dropped class **AtomicSchedule**
- Dropped class **AtomicScheduleValue**
- Added class **AbstractAtomicSchedule**
- Added class **ConstantValueSchedule**
- Added class **TimeSeriesSchedule**
- Class **DualValueSchedule** is now derived from **AbstractAtomicSchedule**
- Class **ScheduleComponent** is now derived from **AbstractADEFeatureWithLifeSpan**

Time series package:

- Added class **AbstractTimeSeriesFile**
- Added class **MonthlyTimeSeries**
- Added class **MonthlyTimeSeriesFile**
- Added class **TypicalValuesMonthlyTimeSeries**
- Added class **TypicalValuesMonthlyTimeSeriesFile**
- Added class **IrregularTimeSeries**
- Added class **IrregularTimeSeriesFile**
- Added class **TypicalValuesIrregularTimeSeries**
- Added class **TypicalValuesIrregularTimeSeriesFile**
- Renamed class **TypicalValuesTimeSeries** to **TypicalValuesRegularTimeSeries**
- Renamed class **TypicalValuesTimeSeriesFile** to **TypicalValuesRegularTimeSeriesFile**
- Class **RegularTimeSeries**:
  - Dropped property *temporalExtent*
  - Added property *startTimestamp*
  - Added property *endTimestamp*
- Class **RegularTimeSeriesFile**:
  - Class is now derived from class **AbstractTimeSeriesFile**
  - Dropped property *temporalExtent*
  - Dropped property *uom*
  - Dropped property *fileURI*
  - Dropped property *numberOfHeaderLines*
  - Dropped property *fieldSeparator*
  - Dropped property *recordSeparator*
  - Dropped property *decimalSymbol*
  - Dropped property *valueColumnNumber*
  - Added property *startTimestamp*
  - Added property *endTimestamp*
- Class **TypicalValuesRegularTimeSeriesFile**:
  - Class is now derived from class **AbstractTimeSeriesFile**
  - Dropped property *uom*
  - Dropped property *fileURI*
  - Dropped property *numberOfHeaderLines*
  - Dropped property *fieldSeparator*
  - Dropped property *recordSeparator*
  - Dropped property *decimalSymbol*
  - Dropped property *valueColumnNumber*

Version 3.0 beta 7

2025-10-14

No changes, except for version number going from 2.0 to 3.0.

Version 2.0 beta 7

2025-06-25

Schedules package:
- Class **ScheduleLibrary**
  - Changed relation type of *libraryMember* to composition

Layered Construction package:
- Class **LayeredConstructionLibrary**
  - Changed relation type of *libraryMember* to composition
- Class **LayeredConstruction**
  - Added property *gValue*
- Class **MaterialLibrary**
  - Changed relation type of *libraryMember* to composition

2025-05-30

Schedules package:
- Class **AtomicSchedule**
  - Changed multiplicity of relation *timeSeries* to [0..1]
  - Added property *constantValue*

2025-05-13

Building physics package:
- Class **ThermalZone**
  - Changed relation type of *thermalBoundary* to composition

2025-05-07

Core package:
- Class **ADE _CityObject**
  - Changed relation type of *device* to aggregation
  - Changed relation type of *weatherData* to composition
- Class **AbstractThermalZone**
  - Changed relation type of *usageZone* to association
- Class **AbstractUsageZone**
  - Changed relation type of *buildingUnit* to association
- Class **AbstractResource**
  - Made relation to **AbstractTimeSeries** explicit and set to composition
- Class **DeviceOperation**
  - Made relation to **AbstractSchedule** explicit and set to aggregation
- Class **WeatherData**
  - Made relation to **AbstractTimeSeries** explicit and set to composition

Occupancy package:
- Class **UsageZone**
  - Changed relation type of *occupiedBy* to aggregation
  - Made all relations to **AbstractSchedule** explicit and set to aggregation
- Class **BuildingUnit**
  - Changed relation type of *occupiedBy* to aggregation
- Class **Occupants**

  - Made relation to **AbstractSchedule** explicit and set to aggregation

Layered Construction package:
- Class **LayeredConstruction**
  - Changed relation type of *layer* to composition
- Class **Layer**
  - Changed relation type of *material* to aggregation

Schedules package:
- Class **ScheduleComponent**
  - Changed relation type of *scheduleComponentMember* to aggregation

2025-05-01

Core package:
- Class **ADE _AbstractSchedule**
  - Moved from package Schedules to Core
- Class **ADE _AbstractTimeSeries**
  - Moved from package Time Series to Core

Occupancy package:
- Class **Occupants**:
  - Added property *averageDietType*
  - Added property *averageIncomeLevel*
  - Added property *averageInstructionLevel*

CodeLists package:
- Added class **DietTypeValue**
- Added class **InstructionLevelValue**
- Added class **IncomeLevelValue**

2025-04-23

CodeLists package:
- Added values to class **AreaTypeValue**
- Renamed class **HeightReferenceValue** to **HeightTypeValue**
- Added class **SuitabilityReasonValue**
- Added class **UrbanSpaceTypeValue**
- Added class **UrbanSpaceEndUseValue**

DataTypes package:
- Added class **Suitability**

Core package:
- Class **ADE _CityObject**
  - Added property *suitability*
- Class **ADE _AbstractBuilding**
  - Renamed property *bdgFloorArea* to *bdgArea*
  - Renamed property *bdgHeightAboveGround* to *bdgHeight*
- Class **AbstractBuildingPartition**
  - Renamed property *floorArea* to *area*

Resources package:
- Added class **UrbanSpace**

2025-04-12

Core package:
- Added class **AbstractLibrary**

Layered Construction package:
- Added class **LayeredConstructionLibrary**
- Added class **MaterialLibrary**

Schedules package:
- Added class **ScheduleLibrary**

CodeLists package:
- Class **FunctionInNetwork** renamed to **FunctionInNetworkValue**

Version 2.0 beta 6

2025-02-06

DataTypes package:
- Dropped class **HeatExchangeType**
- Dropped class **OpticalProperties**

Core package:
- Class ADE _**CityObject:**
  - Renamed property from *operationSchedule* to *deviceOperation*
- Class **AbstractDevice:**
  - Changed type of property *heatDissipation* to gml:Measure
  - Added property *heatDissipationConvectiveFraction*
  - Added property *heatDissipationLatentFraction*
  - Added property *heatDissipationRadiantFraction*

Occupancy package:
- Class **UsageZone**:
  - Renamed property *averageInternalGains* to *internalGains* and changed type to gml:Measure
  - Added property *internalGainsConvectiveFraction*
  - Added property *internalGainsLatentFraction*
  - Added property *internalGainsRadiantFraction*
- Class **Occupants**:
  - Changed type of property *heatDissipation* to gml:Measure
  - Added property *heatDissipationConvectiveFraction*
  - Added property *heatDissipationLatentFraction*
  - Added property *heatDissipationRadiantFraction*

Layered Construction package:
- Class **LayeredConstruction**:
  - Dropped property *opticalProperties*
  - Added property *glazingRatio*
  - Added property *emissivity*
  - Added property *reflectance*
  - Added property *transmittance*
- Class **Layer**:
  - Changed multiplicity of property *thickness* to [1]
- Class **AbstractMaterial**:
  - Added property *libraryCode*

Version 2.0 beta 5

Resources package:

No changes since beta 4

Weather Station package:
No changes since beta 4

2025-01-30

Packages
- Created TimeSeries package and moved all related classes into it
- Created Schedule package and moved all related classes into it
- Dropped Supporting classes package

CodeLists package:
- Class **CertificateTypeValue** renamed to **EnergyPerformanceCertificateTypeValue**

2025-01-28

CodeLists package:
- Added class **ScheduleTypeValue**

Enumerations package:
- Dropped class **DayTypeValue**

Supporting classes package:
- Dropped class **TimeSeriesSchedule**
- Dropped class **DailyPatternSchedule**
- Dropped class **PeriodOfYear**
- Dropped class **DailySchedule**
- Added class **AtomicSchedule**
- Added class **CompositeSchedule**
- Added class **ScheduleComponent**
- Class **AbstractSchedule**:
  - Added properties *type*, *startTime*, *startDay*, *startMonth*, *startYear*, *temporalExtent*

2025-01-22

Devices package:
- Class **AbstractSolarCollector**:
  - Added properties *azimuth* and *inclination*
  - Dropped property *surfaceGeometry*
  - Added properties *lod2MultiSurface* and *lod3MultiSurface*

2025-01-13

CodeLists package:
- Dropped class **ThermalBoundaryTypeValue**
- Added class **ShadingDeviceTypeValue**
- Added class **WeatherValueTypeValue**

DataTypes package:
- Moved class **PeriodOfYear** from package DataTypes to Supporting classes
- Moved class **DailySchedule** from package DataTypes to Supporting classes
- Class **AbstractQualifiedAttribute**:
  - Added properties *source* and *value*
- Classes **QualifiedHeight, QualifiedArea, QualifiedVolume**:
  - Dropped property *value*

Supporting classes package:
- Class **AbstractTimeSeries**:

  - Moved back from package Core to Supporting classes
  - Changed multiplicity of property *acquisitionMethod* to [1]
  - Changed multiplicity of property *interpolationType* to [1]
- Class**TypicalValuesTimeSeries**:
  - Added properties *startTime*, *startDay*, *startMonth*
- Class **TypicalValuesTimeSeriesFile**:
  - Added properties *startTime*, *startDay*, *startMonth*
- Added class **DualValueSchedule**:
- Class **PeriodOfYear**:
  - Changed stereotype from dataType to featureType
  - Added properties *startDay*, *startMonth*, *endDay*, *endMonth*
  - Dropped property *period*
- Class **DailySchedule**:
  - Changed stereotype from dataType to featureType
  - Added property *repetition*
  - Renamed property from *schedule* to *timeSeries*

Layered Construction package:

- Added class **ReverseLayeredConstruction**
- Dropped class **LayerComponent**
- Class **Layer**:
  - Removed property *layerComponent*
  - Added property *thickness*
  - Added property *material*
- Class **SolidMaterial**:
  - Renamed property from *specificHeat* to *specificHeatCapacity*
  - (Re)added and renamed property from *conductivity* to *thermalConductivity*
  - (Re)added property *embodiedEnergy*
  - (Re)added property *embodiedCarbon*
  - (Re)added property *porosity*

Core package:

- Class **AbstractDevice**:
  - Class is renamed from **AbstractEnergySystem** to **AbstractDevice**
  - Added property *heatDissipation*
  - Added property *operationSchedule*
  - Added property *efficienctIndicator*
  - Added property *installedPower*
  - Added property *nominalEfficiency*
- Class **AbstractTimeSeries**:
  - Moved back from package Core to Supporting classes
- Class (ADE) **_CityObject**:
  - Renamed property from *energySystem* to *device*
- Class **OperationSchedule**:
  - Moved from package Devices to Core
- Class **WeatherData**:
  - Renamed property from *timeseries* to *timeDependentValues*
  - Renamed property from *singleValue* to *yearlyValue*
  - Renamed property from *singleValueType* to *valueType*
  - Changed type of property valueType from CharacterString to CodeList
- Added class **AbstractBuildingPartition**
- Classes **AbstractBuildingUnit**, **AbstractUsageZone**, **AbstractThermalZone**
  - Changed parent class

Building physics package:

- Dropped class **ThermalBoundary**
- Dropped class **ThermalOpening**
- Class **ThermalZone**
  - Removed property *floorArea* (now inherited via **AbstractBuildingPartition**)

  - Removed property *volume* (now inherited via **AbstractBuildingPartition**)
  - Removed property *volumeGeometry* (now inherited via **AbstractBuildingPartition**)
  - Removed property *boundedBy*
  - Added property *heatCapacity*
  - Added property *coincidesWithLod2Hull*
  - Added property *coincidesWithLod3Hull*
  - Added property *thermalBoundary*
- Added class **PartyWallSurface**
- Added class **BasementCeilingSurface**
- Added class **IntermediateFloorSurface**
- Added class **AtticFloorSurface**
- Class (ADE) **_BoundarySurface**:
  - Renamed property from *bdgBdrySrfWindowToWallRatio* to *bdgBdrySrfOpeningToSurfaceRatio*
  - Renamed property from *bdgBdrySrfArea* to *bdgBdryTotalSurfaceArea*
  - Added property *bdgBdryOpaqueSurfaceArea*
  - Added property *bdgBdrySrfThickness*
  - Added property *bdgBdrySrfHeatCapacity*

Occupancy package:
- Dropped class **AbstractFacilities**
- Dropped class **DHWFacilities**
- Dropped class **LightingFacilities**
- Dropped class **ElectricalAppliances**
- Dropped class **GenericFacilities**
- Class **BuildingUnit**:
  - Removed property *floorArea* (now inherited via **AbstractBuildingPartition**)
  - Removed property *volume* (now inherited via **AbstractBuildingPartition**)
  - Removed property *equippedWith*
- Class **UsageZone**:
  - Removed property *floorArea* (now inherited via **AbstractBuildingPartition**)
  - Removed property *volume* (now inherited via **AbstractBuildingPartition**)
  - Removed property *volumeGeometry* (now inherited via **AbstractBuildingPartition**)
  - Removed property *equippedWith*

Devices package:

The former EnergySystems package is renamed to Devices package
- Class **OperationSchedule**:
  - Moved from package Devices to Core
- Dropped class **AbstractEnergyConversionSystem**
  - Now all subclasses are derived directly from **AbstractDevice**
- Dropped class **AbstractStorageSystem**
  - Now all subclasses are derived directly from **AbstractDevice**
- Class **GenericDevice**:
  - Renamed from class **GenericEnergyConversionSystem**
- Class **ElectricalStorageDevice**:
  - Renamed from class **PowerStorageSystem**
- Class **ThermalStorageDevice**:
  - Renamed from class **ThermalStorageSystem**
- Class **AbstractSolarCollector**:
  - Renamed from class **AbstractSolarEnergySystem**
- Class **PhotovoltaicThermalCollector**:
  - Renamed from class **PhotovoltaicThermalSystem**
- Class **PhotovoltaicCollector**:
  - Renamed from class **PhotovoltaicSystem**
- Class **SolarThermalCollector**:
  - Renamed from class **SolarThermalSystem**
- Class **GenericSolarCollector**:
  - Renamed from class **GenericSolarEnergySystem**
- Added class **LightingDevice**

- Added class **GenericElectricalDevice**
- Added class **MovableShadingDevice**

Urban Function Areas package:

- Class **UrbanFunctionArea**:
  - Removed property *rank*

Version 2.0 beta 4

2024-09-25

- Codelist **ThermalBoundaryTypeValue**
  - Renamed *outerWall* to *exteriorWall*
  - Renamed *intermediaryFloor* to *intermediateFloor*

2024-09-20

- Minor edits to the layout of the UML diagrams
- Polished Codelist **ThermalBoundaryTypeValue**

2024-08-12

- Class **ADE CityObject**:
  - Renamed association *connectedTo* to *utilityNetworkConnection*
  - Remodelled association class **CityObjectRelation**
- Class **AbstractTimeSeries**:
  - Moved to Core module
- Class **WeatherData**:
  - renamed property *valuesList* to *timeSeries*
  - made the association to **AbstractTimeSeries** explicit
- Added some values to codelist **RelationTypeValue**
- Class **TimeSeriesSchedule**:
  - Changed association to **AbstractTimeSeries** to Composition
- Class **DailySchedule**:
  - Changed association to **AbstractTimeSeries** to Composition
- Class **SensorConnection**:
  - Changed type of *uom* to UnitOfMeasure
  - Renamed *linkTosensortDescription* to *linkToSensorDescription*
- Class **AbstractSolarEnergySystem**:
  - Removed association *installedOn* (now taken over by **CityObjectRelation**)
- Class **Energy**:
  - Changed multiplicity of *EnergyCarrier* to [0..1]
- Class **Waste**:
  - Changed multiplicity of *isDanegerous* and *isRecyclable* to [0..1]

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